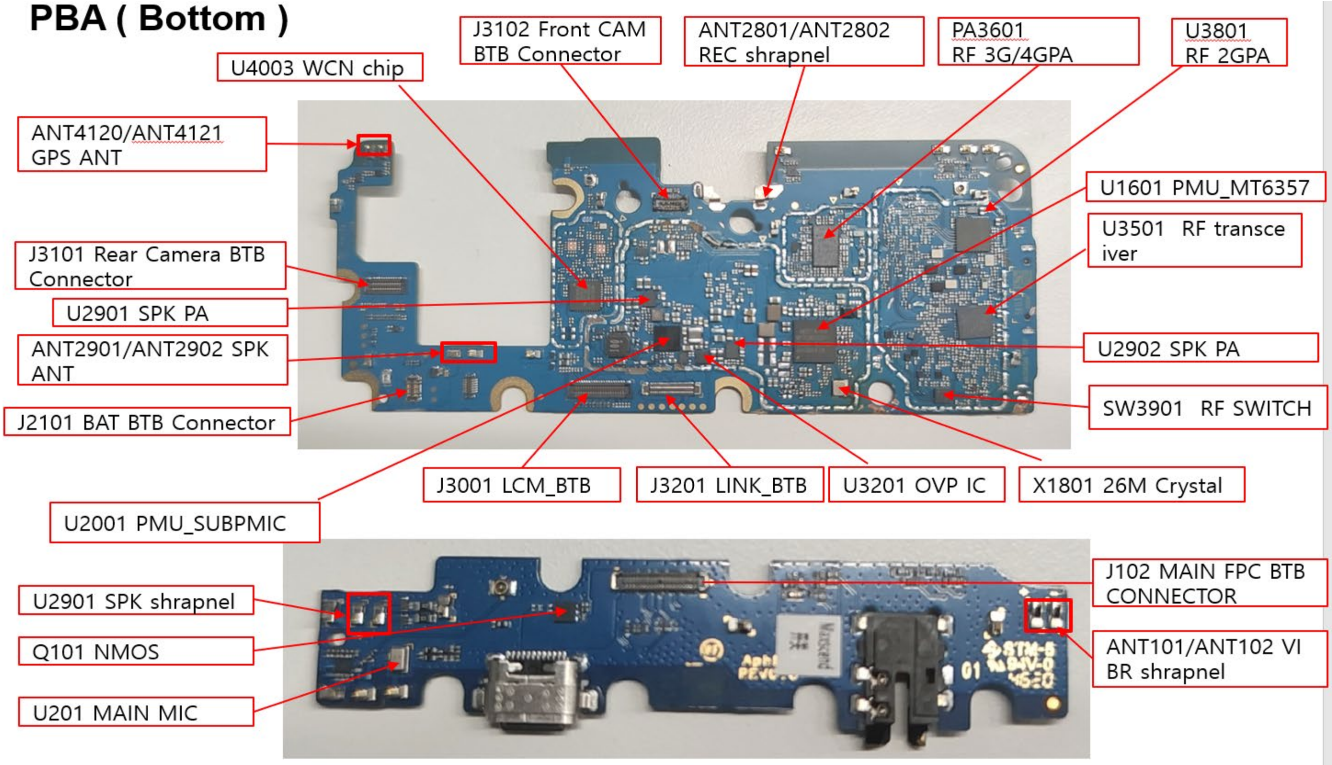
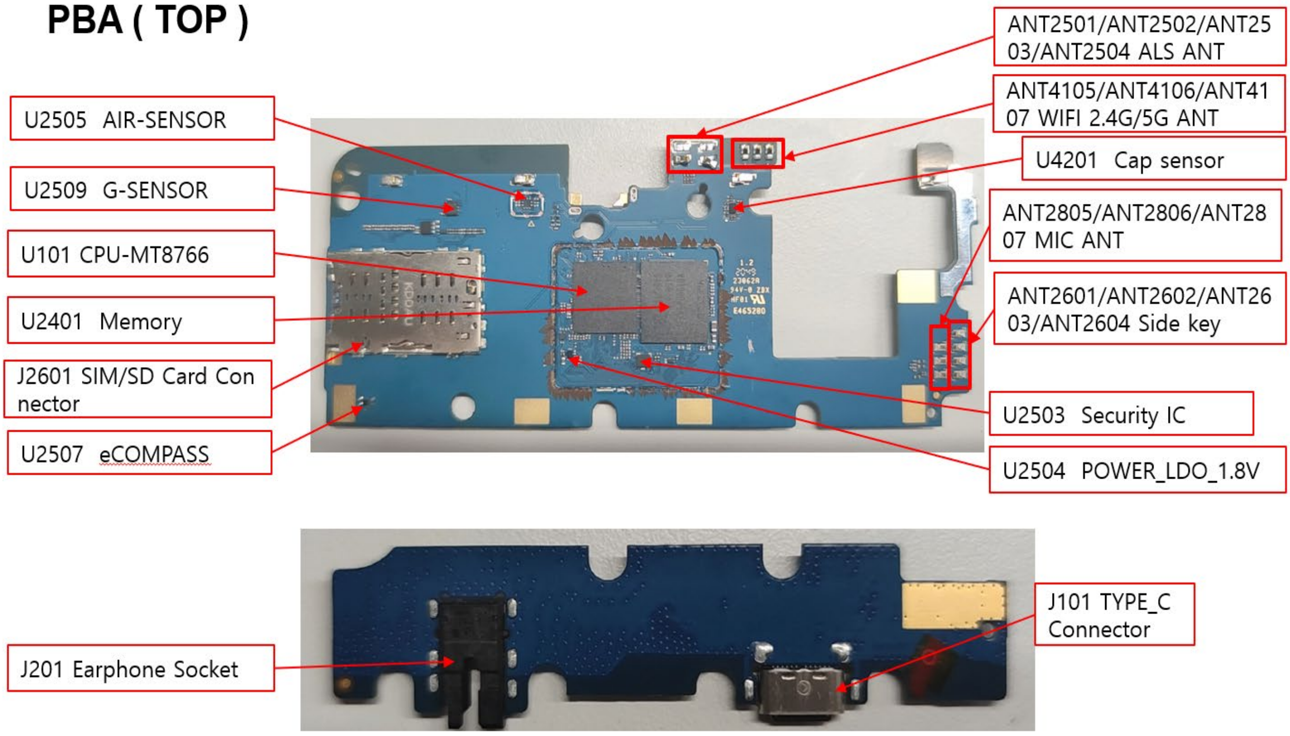
8. Level 3 Repair

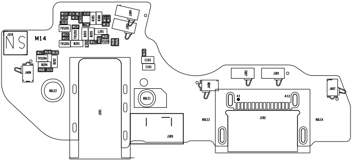
8-1. Components Layout

PBA (Bottom)

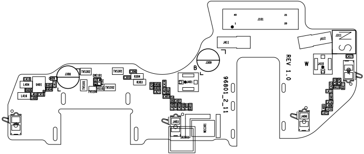


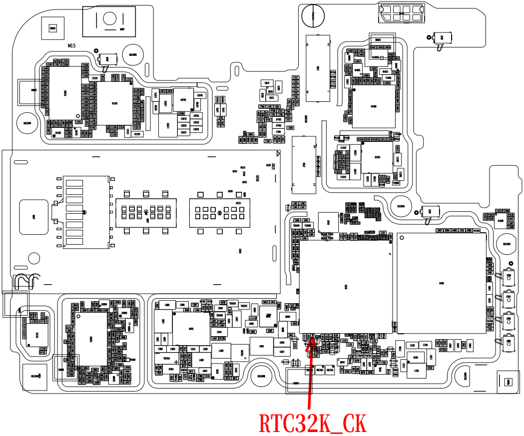
PBA (TOP)

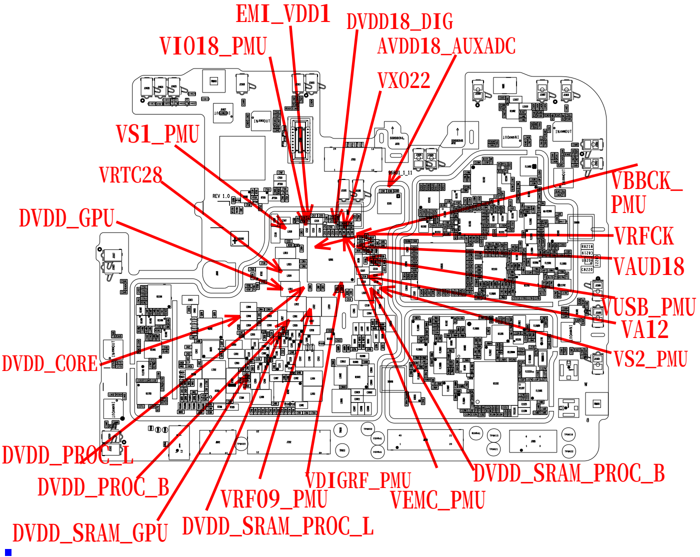




J310
M14
J409
HOLE2
J201
HOLE1
J305
HOLE3
J302
J301
A1
A12
HOLE4
J407
C103
C101
TVS201
TVS204
L201

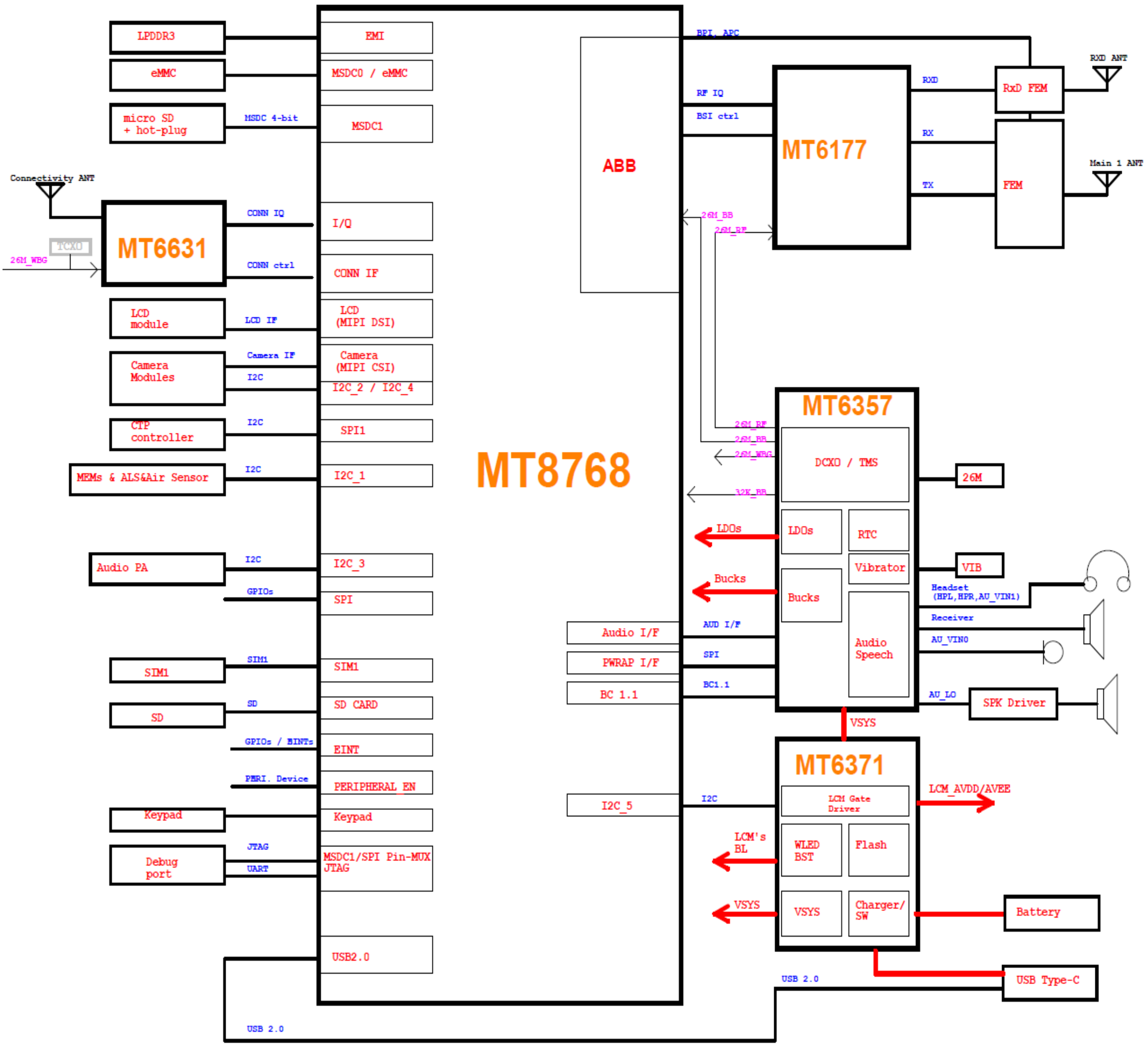

J101
J411
J412
J311
J309
J308
J401
J403
J406
J410
U401
L404
L416
TVS301
TVS302
R304
R303
MIC301
B
W
REV 1.0
96801_2_11

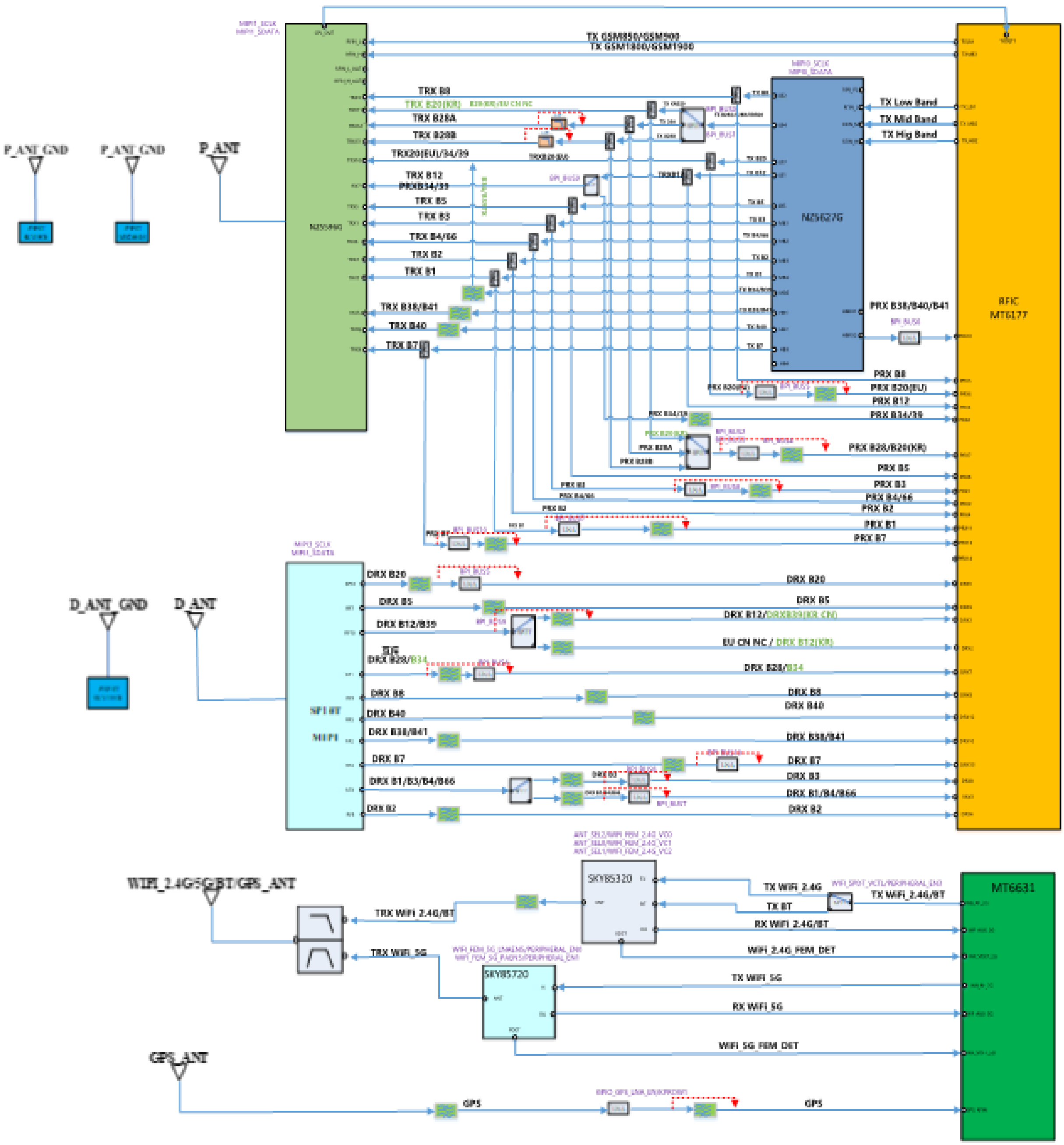

M15
RTC32K_CK



EMI_VDD1
DVDD18_DIG
VIO18_PMU
AVDD18_AUXADC
VX022
VS1_PMU
VRTC28
VBBCK_PMU
DVDD_GPU
VRFCK
VAUD18
VUSB_PMU
VA12
VS2_PMU
DVDD_CORE
DVDD_PROC_L
DVDD_PROC_B
VDIGRF_PMU
VEMC_PMU
DVDD_SRAM_PROC_B
VRF09_PMU
DVDD_SRAM_GPU
DVDD_SRAM_PROC_L

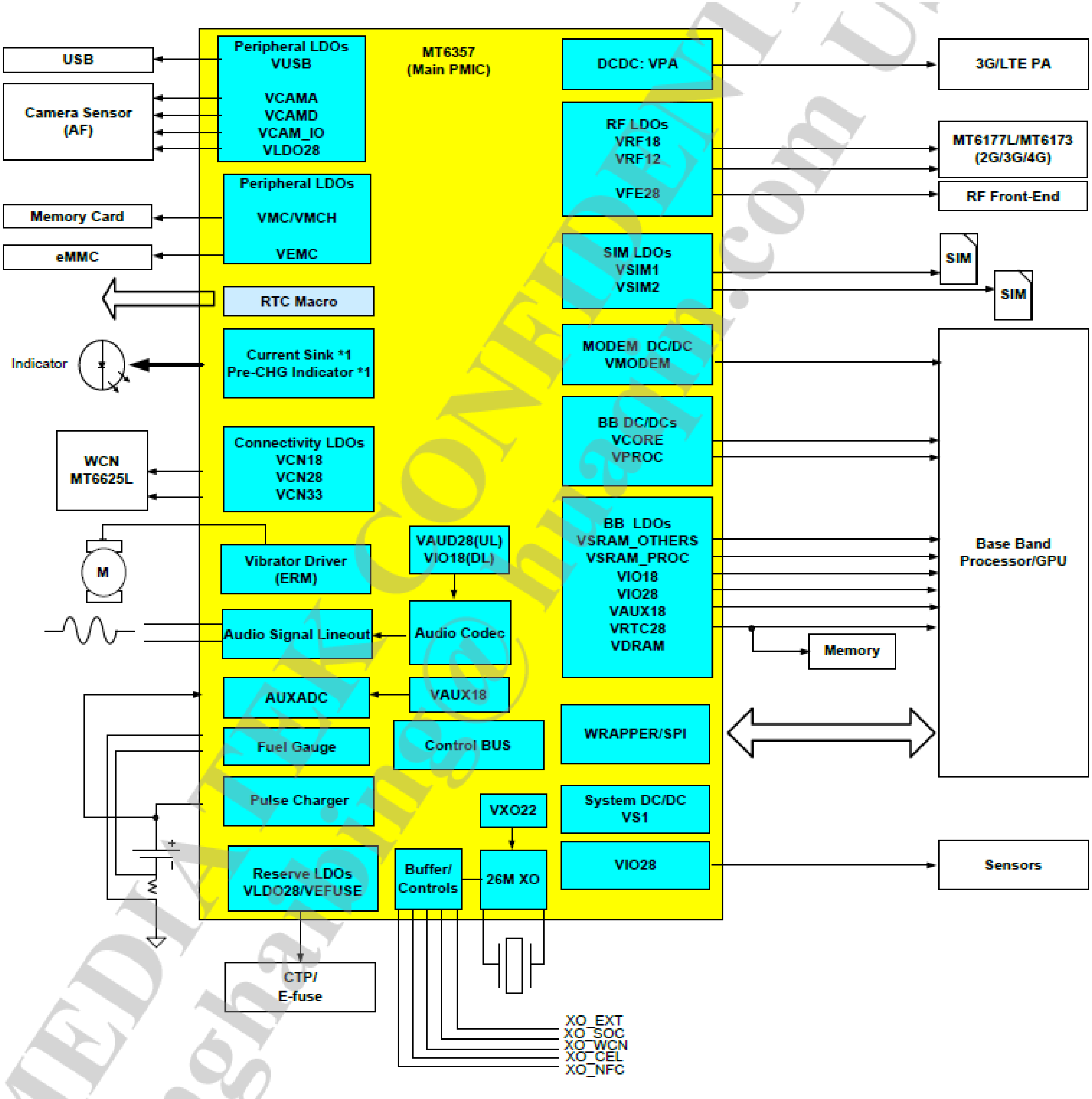
8. Level 3 Repair





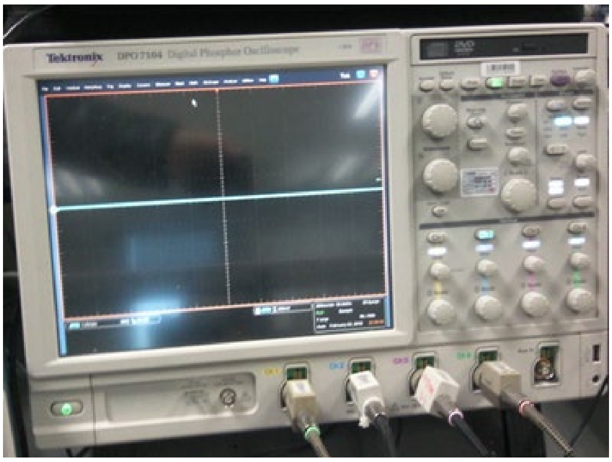
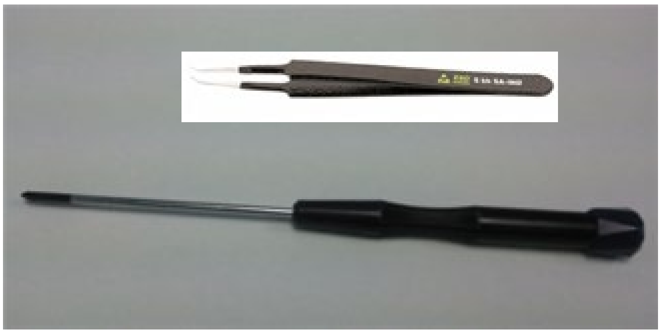
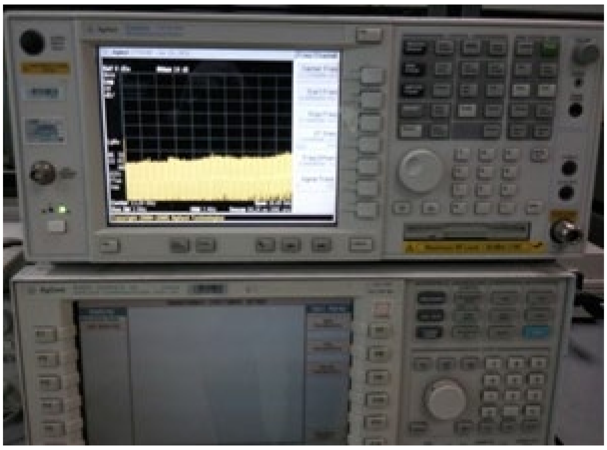
P_ANT_GND
P_ANT_GND
P_ANT
TX GSM850/GSM900
TX GSM1800/GSM1900
TRX B8
TRX B28A
TRX B28B
TRX20(EU)/34/39
TRX B12
PRXB34/39
TRX B5
TRX B3
TRX B4/66
TRX B2
TRX B1
TRX B38/B41
TRX B40
TRX B7
NZ5627G
TX Low Band
TX Mid Band
TX Hig Band
PRX B38/B40/B41
RFIC
MT6177
PRX B8
PRX B20(EU)
PRX B12
PRX B34/39
PRX B28/B20(KR)
PRX B5
PRX B3
PRX B4/66
PRX B2
PRX B1
PRX B7
D_ANT_GND
D_ANT
DRX B20
DRX B5
DRX B12/B39
DRX B28/B34
DRX B8
DRX B40
DRX B38/B41
DRX B7
DRX B1/B3/B4/B66
DRX B2
DRX B12/DRXB39(KR CN)
EU CN NC / DRX B12(KR)
DRX B28/B34
DRX B8
DRX B40
DRX B38/B41
DRX B7
DRX B3
DRX B1/B4/B66
DRX B2
WIFI_2.4G/5G/BT/GPS_ANT
TRX WIFI_2.4G/BT
TRX WIFI_5G
SKY85320
SKY85720
TX WIFI_2.4G
TX BT
TX WIFI_2.4G/BT
RX WIFI_2.4G/BT
WIFI_2.4G_FEM_DET
TX WIFI_5G
RX WIFI_5G
WIFI_5G_FEM_DET
MT6631
GPS_ANT
GPS
GPS

8. Level 3 Repair



8. Level 3 Repair

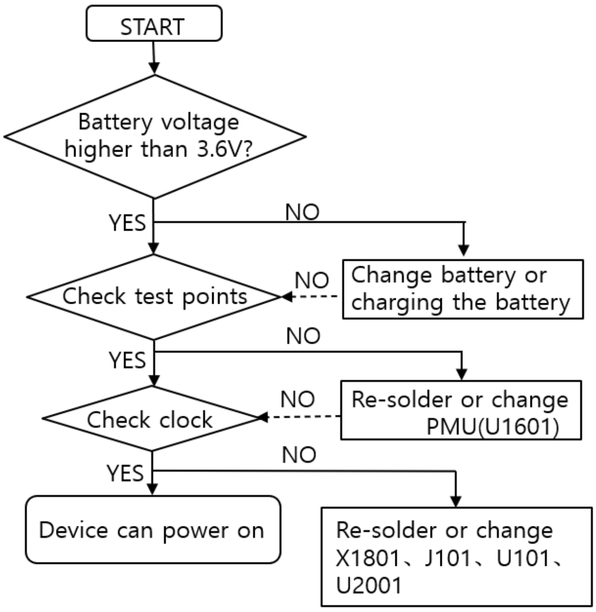
8-3. Flow chart of Troubleshooting.

	
Oscilloscope	**Digital Multimeter**
	
Power Supply	**+ driver, ESD Safe Tweezer**
	
8960 & Spectrum Analyzer	**Soldering iron**

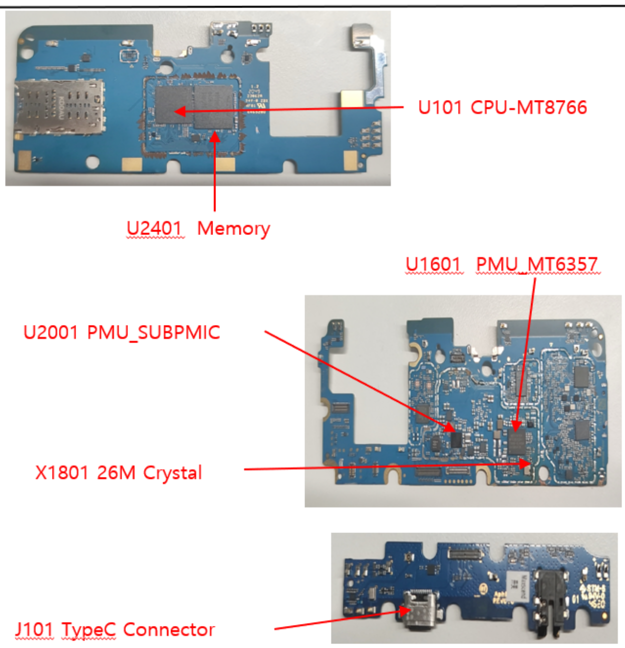
8-3-1. Power On

■ Checking Power signal (Battery connector, PMU, Clock)

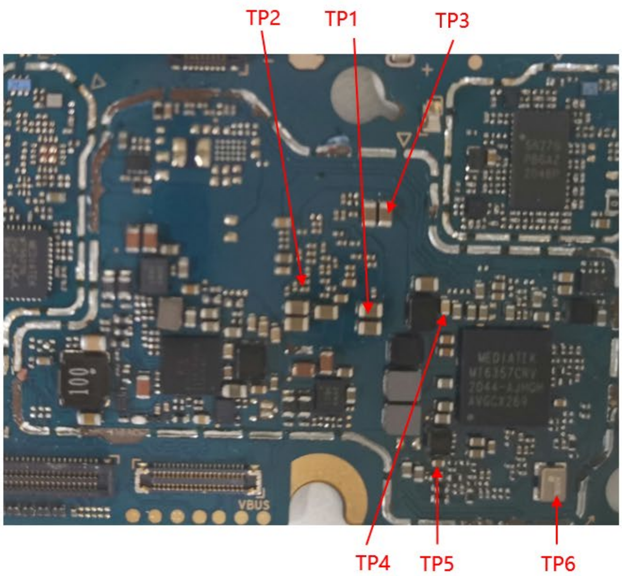
CHECKING FLOW



IMAGE



Power on voltage check				
Power Domain	Configurable Voltage	Signal Name	Measurment Location	TP
VDVFS	0.51875-1.31V	DVDD_DVFS	C1012	TP1
VCORE	0.51875-1.2V	DVDD_CORE	C1010	TP2
VMODEM	0.65-1.2V	DVDD_MODEM	C1009	TP3
VS1	2V	VS1_PMU	C1701	TP4
VPA	0.5V	VPA_PMU	C1606	TP5



➢ **Oscillate frequency measurement**

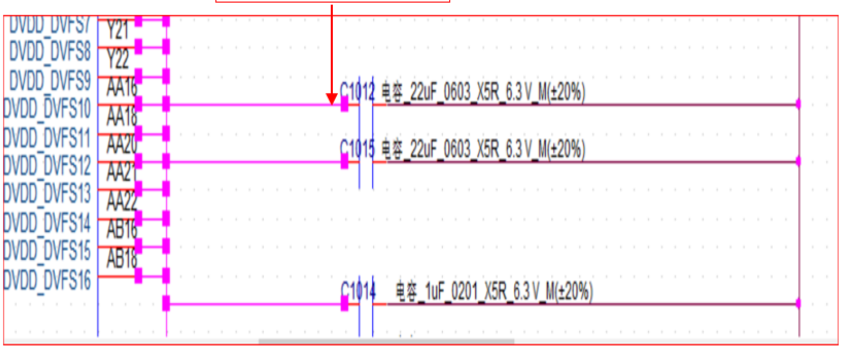
Signal Name	Frequency MHZ	Measurment Location	TP
AUX_ADC_VIN	26M	X1801	TP6

8. Level 3 Repair

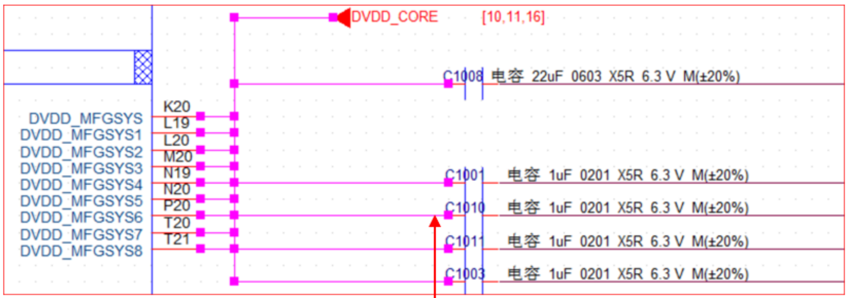
➢ **Checking Power signal(Battery connector, PMU, Clock)**

TP1 C1012 100ohm
0.51875-1.31V

DVDD_DVFS



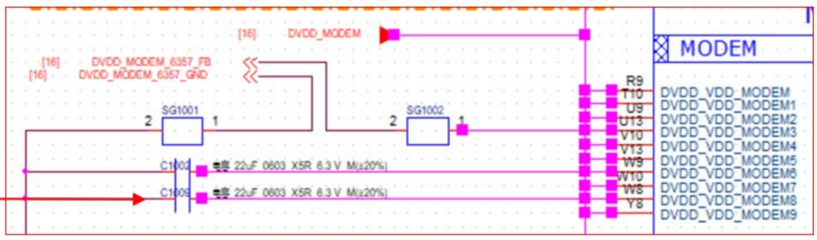
DVDD_CORE



TP2 C1010
0.51875-1.2V 45ohm

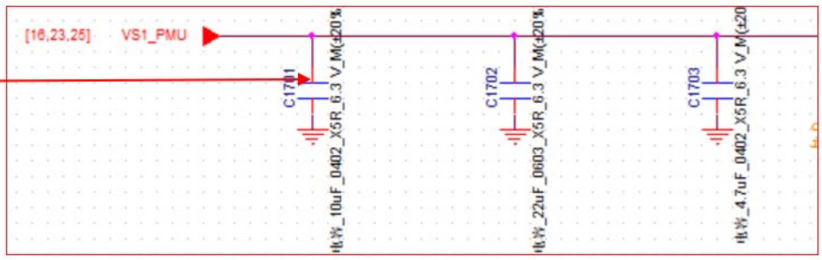
DVDD_MODEM

TP3 C1009
0.65-1.2V
510ohm



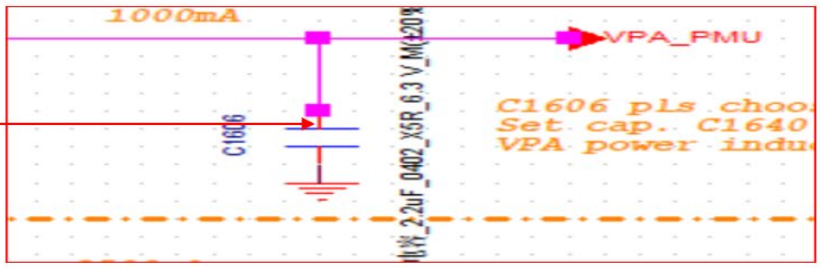
VS1_PMU

TP4 C1701
2V 214Kohm

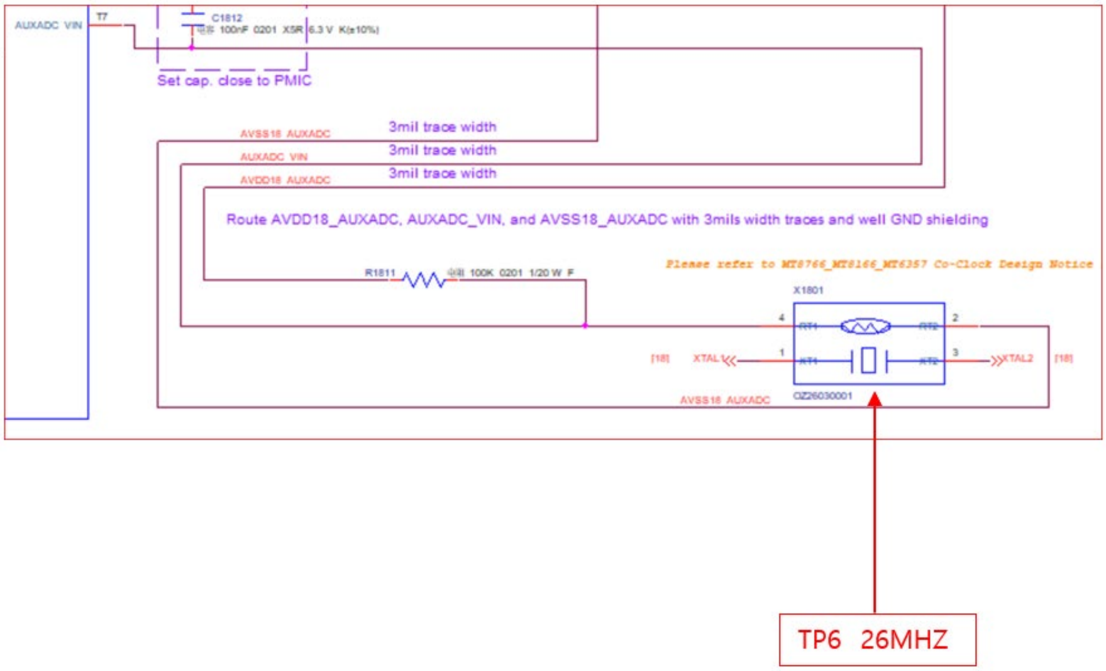


VPA_PMU

TP5 C1606
0.5V 200K

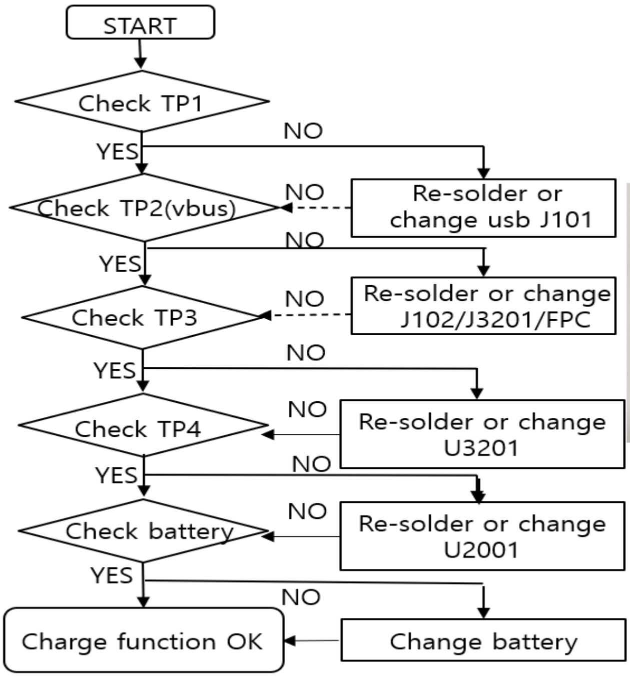
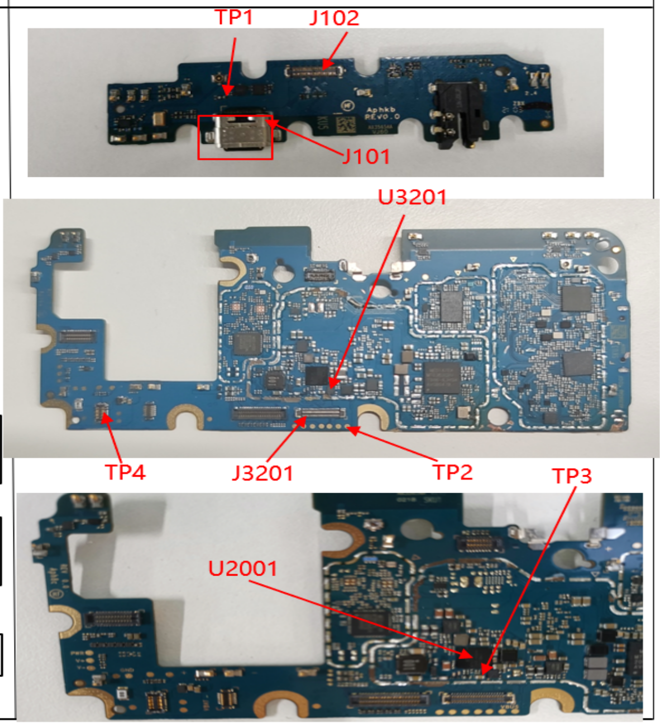


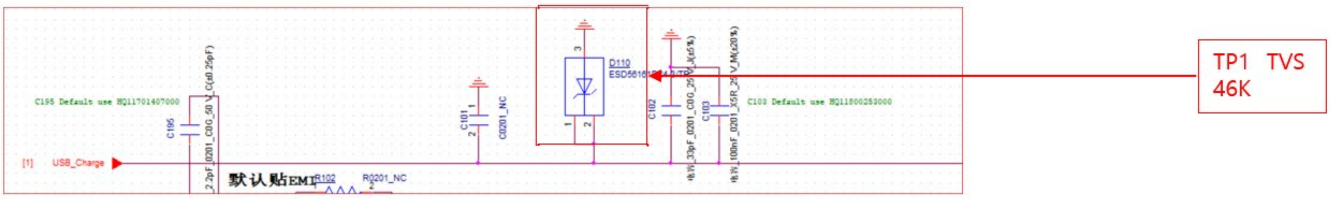
XTAL 26M_IN



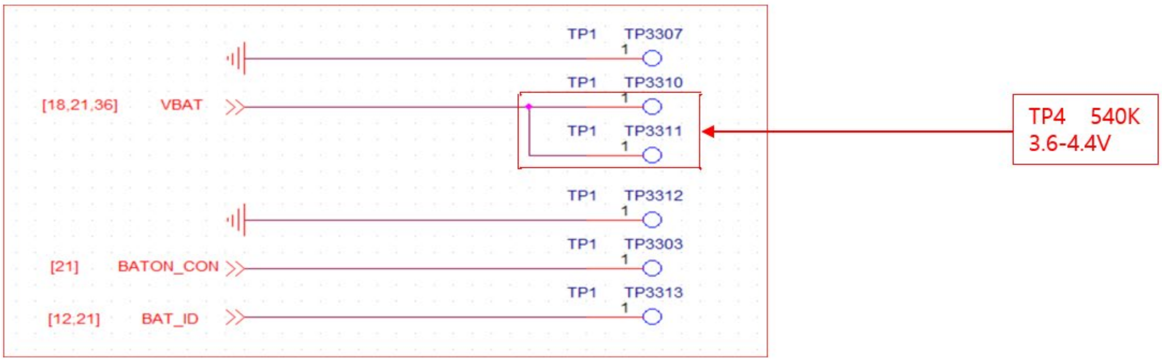
8-3-2. Charging

■ The charging controlled by PMU chip PMI632 (U3201)

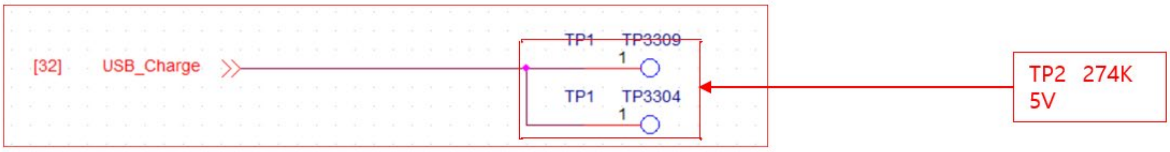
CHECKING FLOW	IMAGE
	



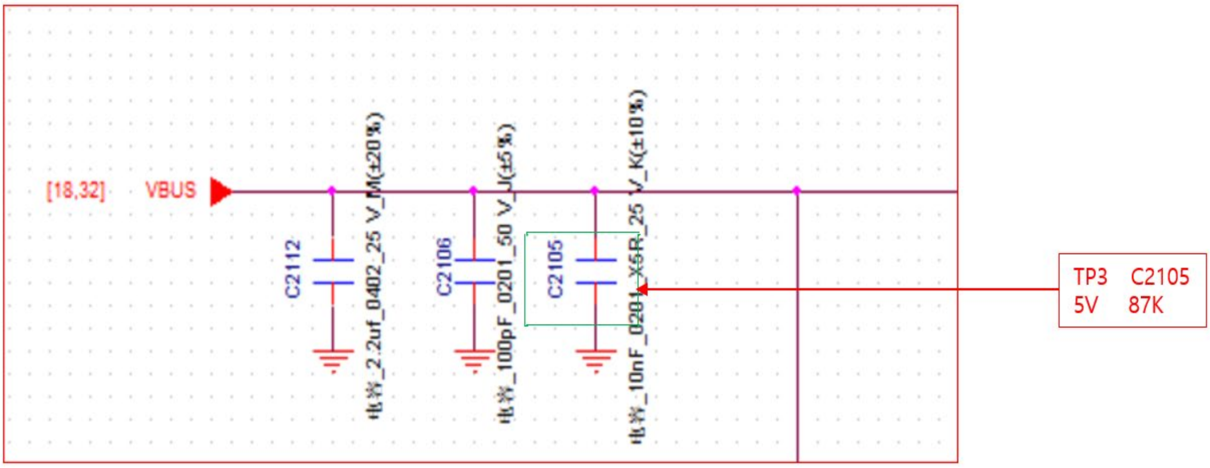
VBAT



VBUS



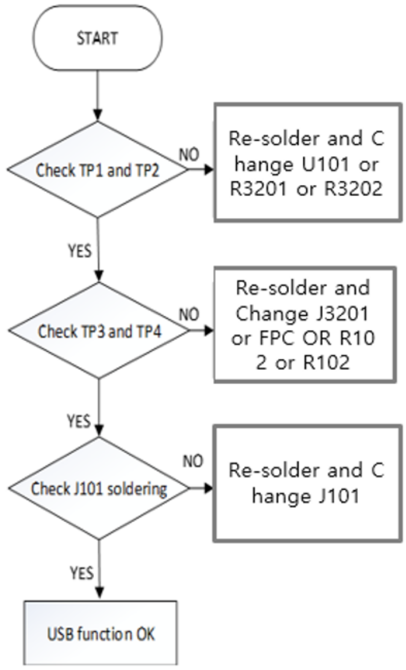
VBUS



8. Level 3 Repair

8-3-3. USB

■ I/O connector is used as the USB port.

<table>
<tr><th>CHECKING FLOW</th><th>IMAGE</th></tr>
<tr><td>


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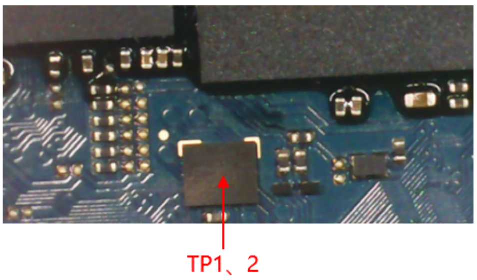

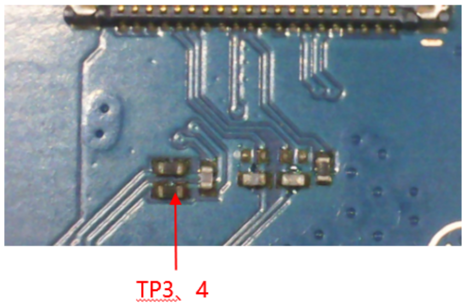

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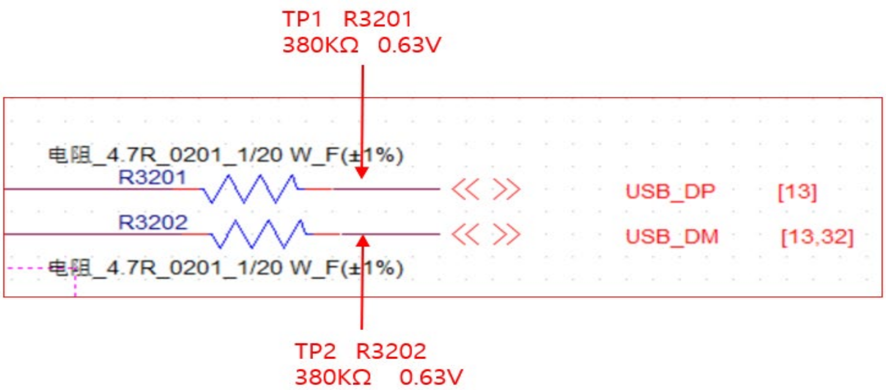
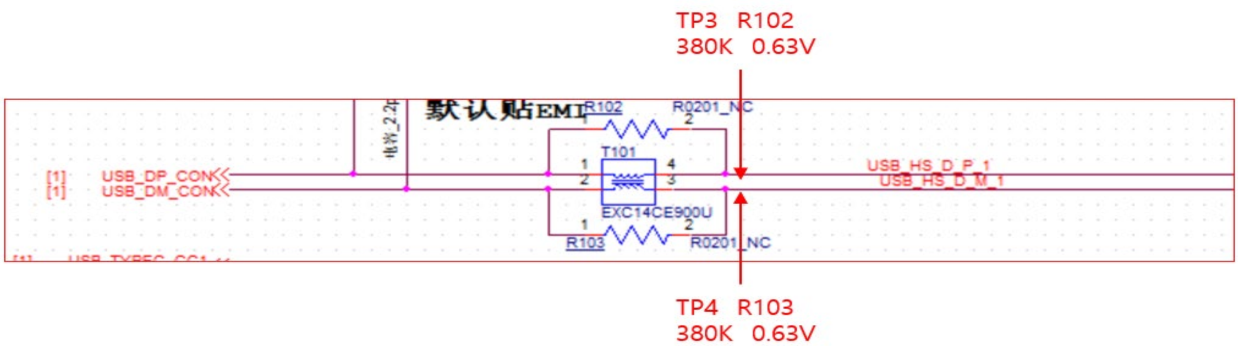
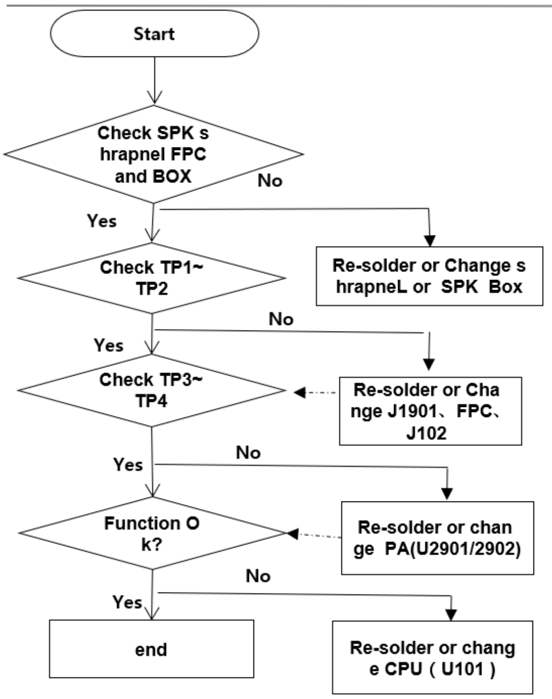


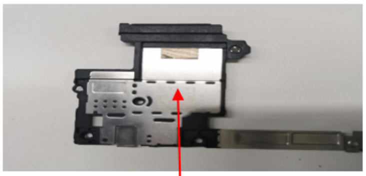


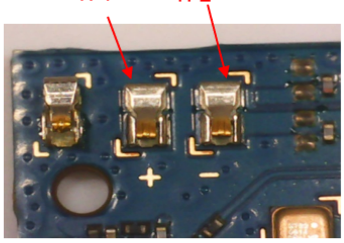
8-3-4. Audio speaker

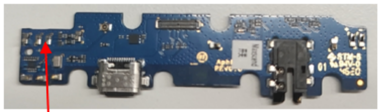
■ The Speaker control signals are generated by MT8766(U101) , There is SPK PA(U2901/U2902) in the middle, the chips and the speaker are to be checked out.

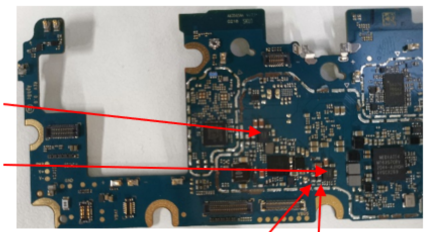
Checking Flow	Image

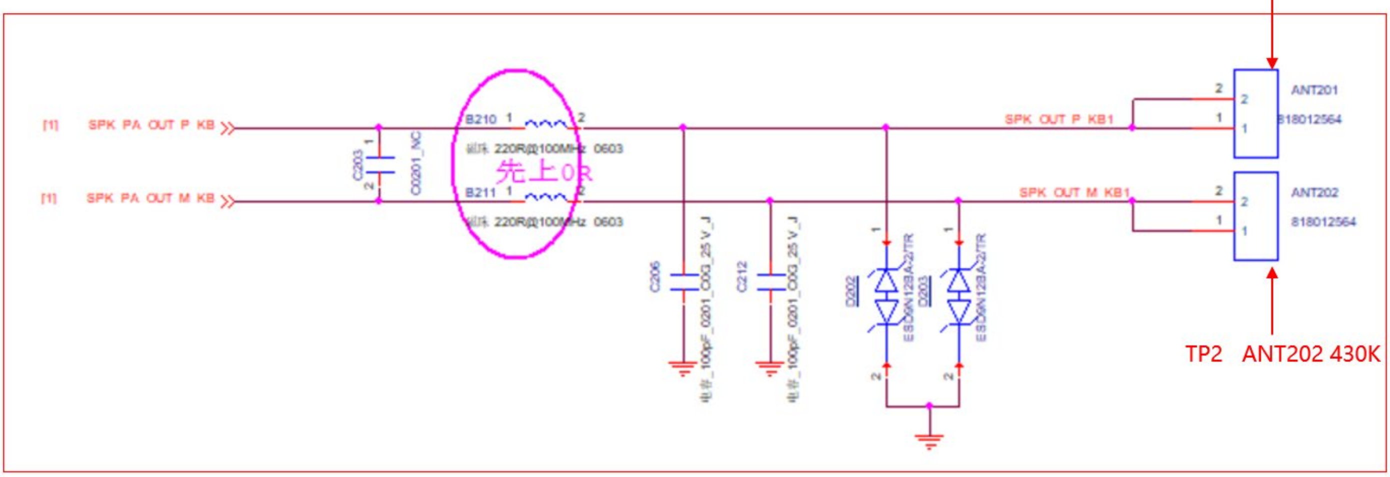


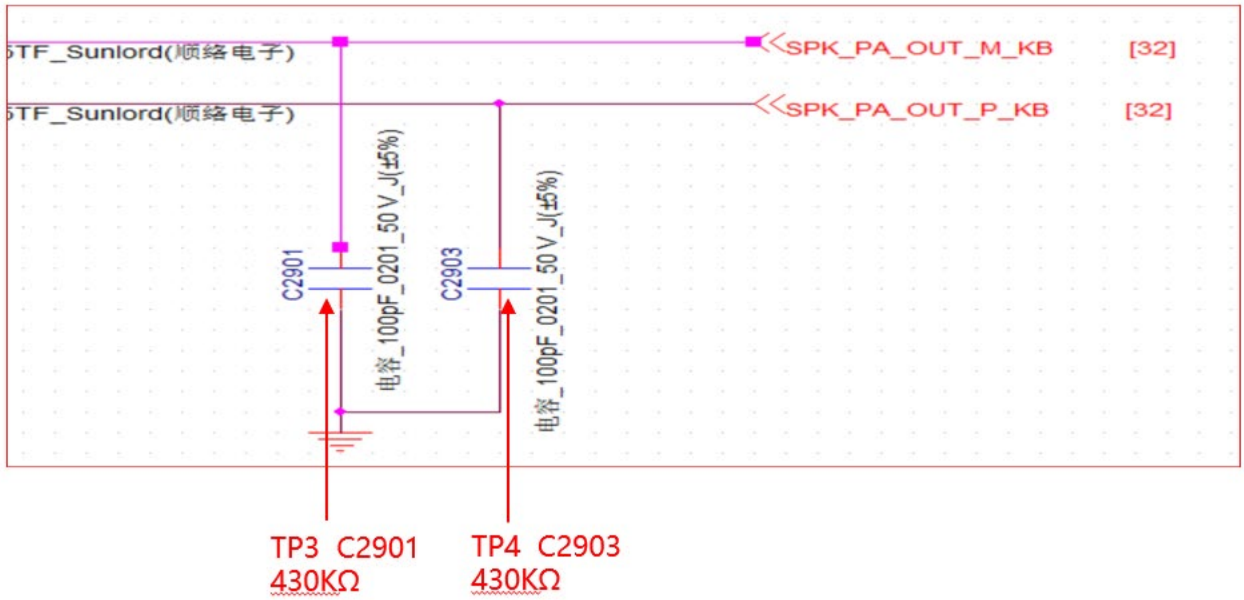








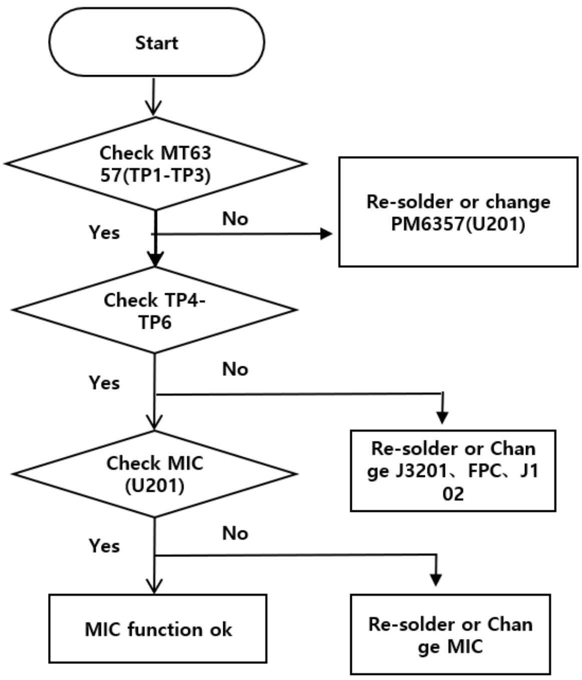


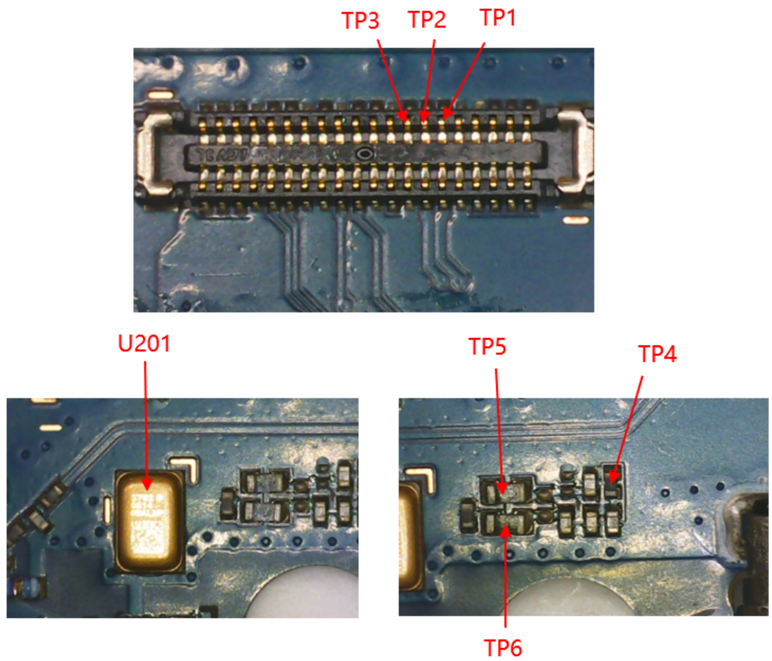


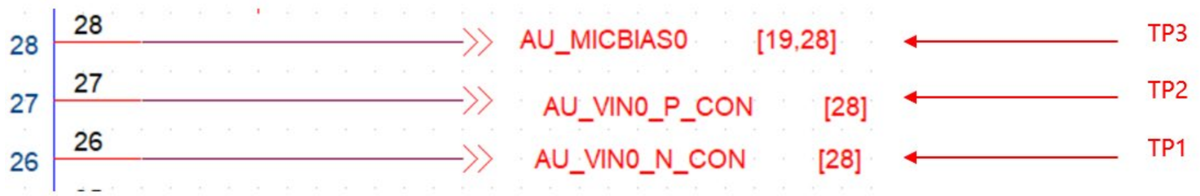
8-3-5. Audio_ Main MIC

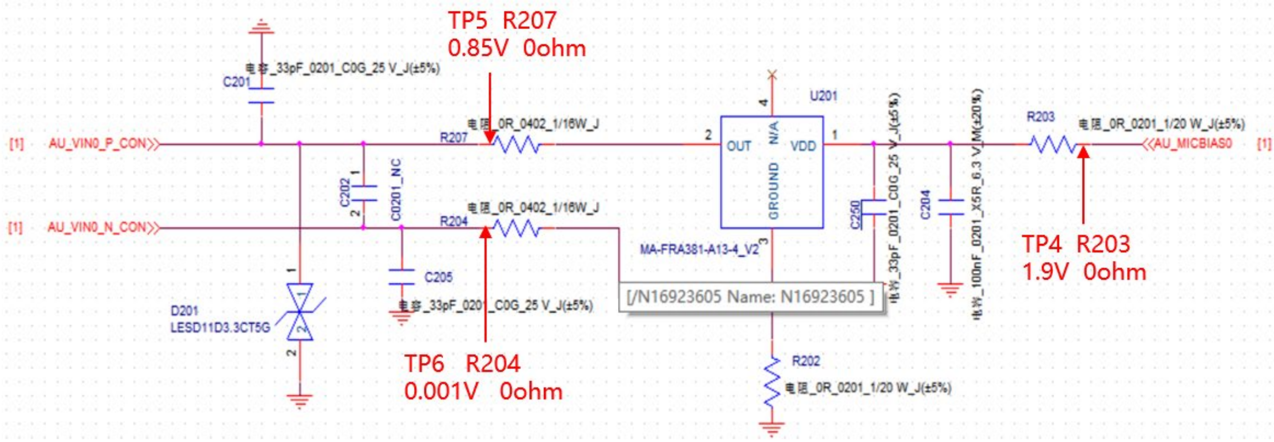
- The MIC control signals are generated by PMU chip MT6357(U1601), the PMU chip and the MIC are to be checked out.

Checking Flow	Image





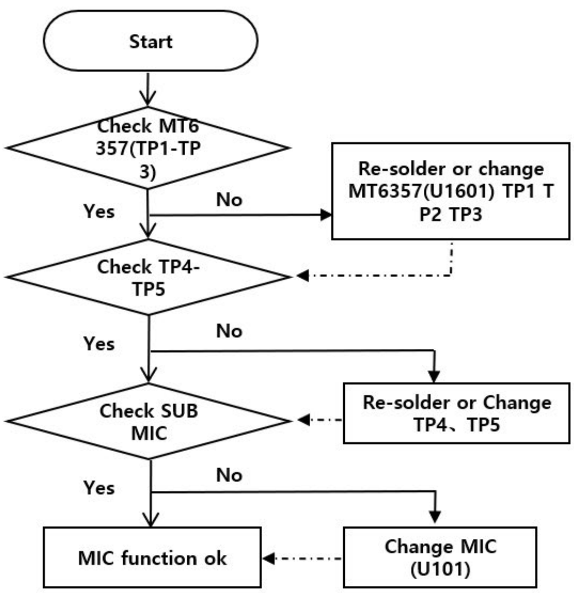




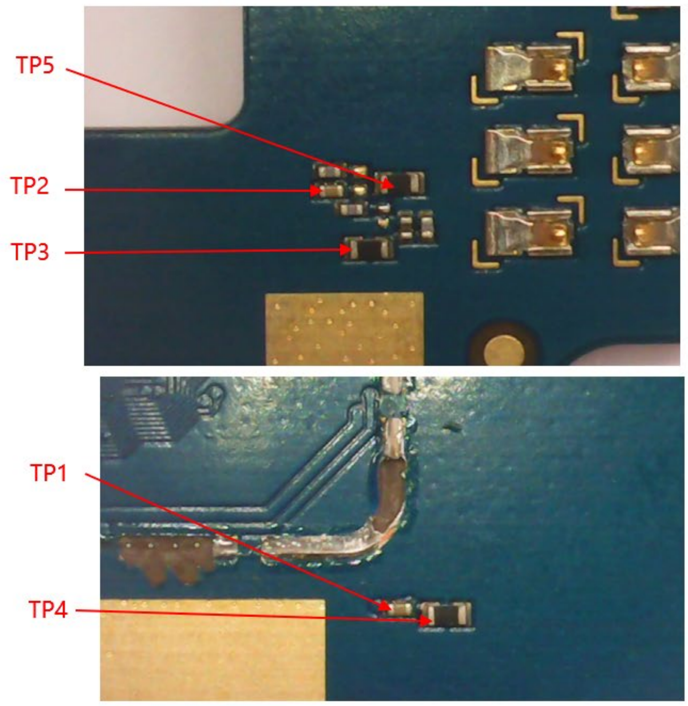
8-3-6. Audio_ Sub MIC

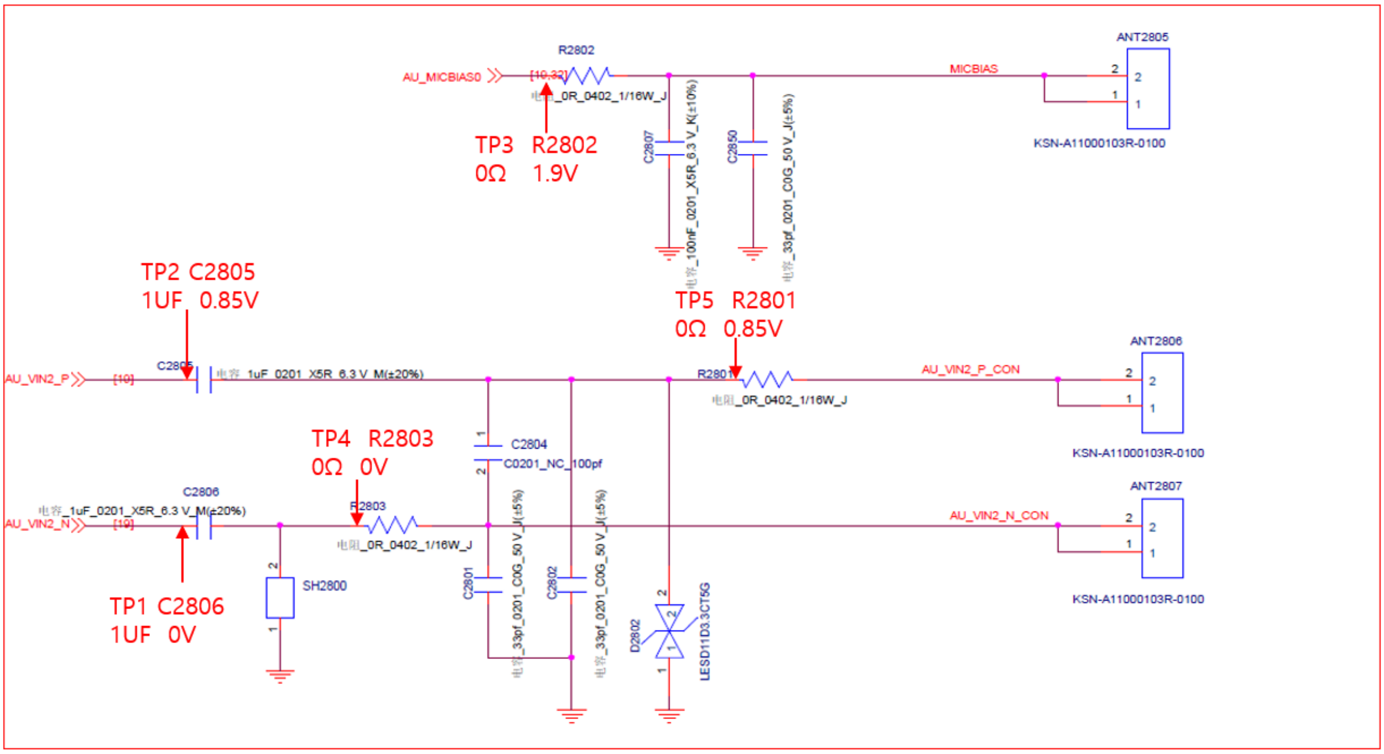
- The sub MIC control signals are generated by PMU chip MT6357(U1601), the PMU chip and the MIC are to be checked out.

Checking Flow



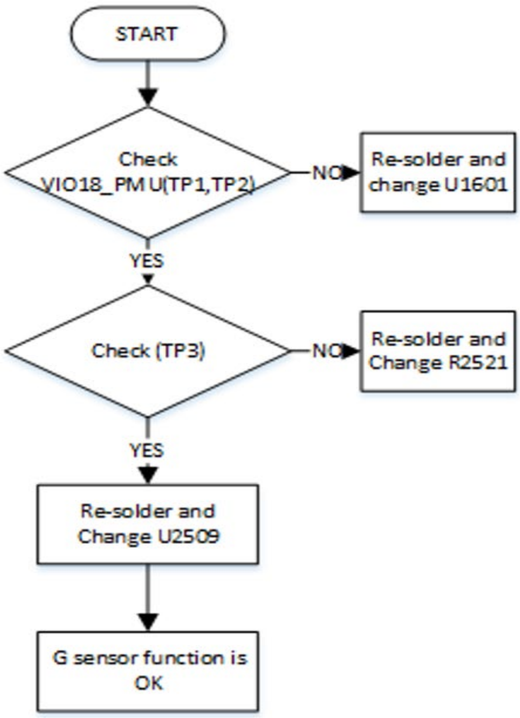
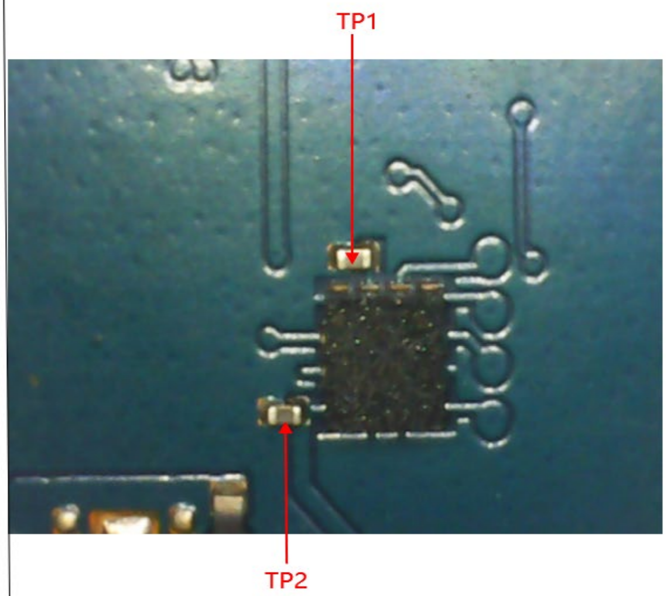
Image

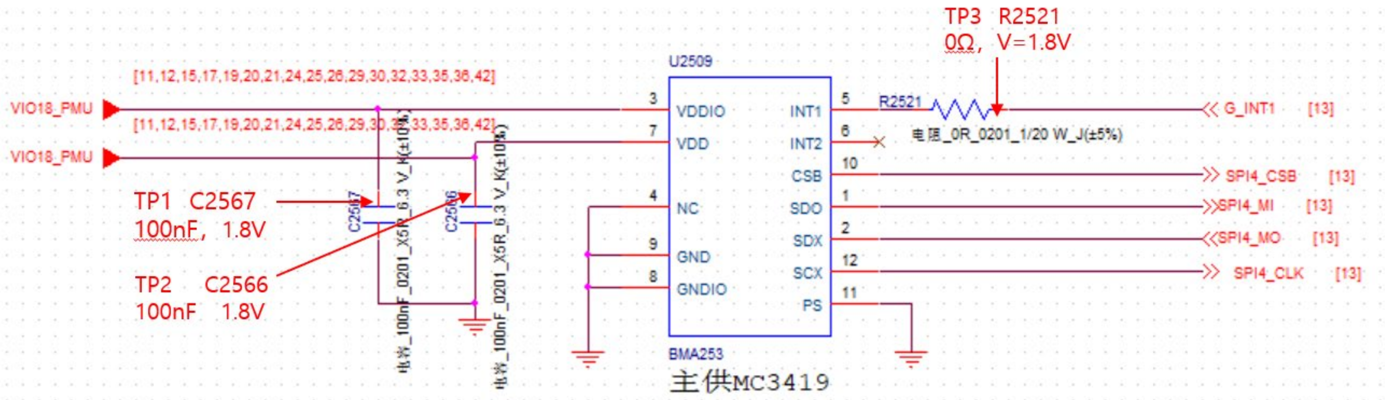




8-3-7. G sensor

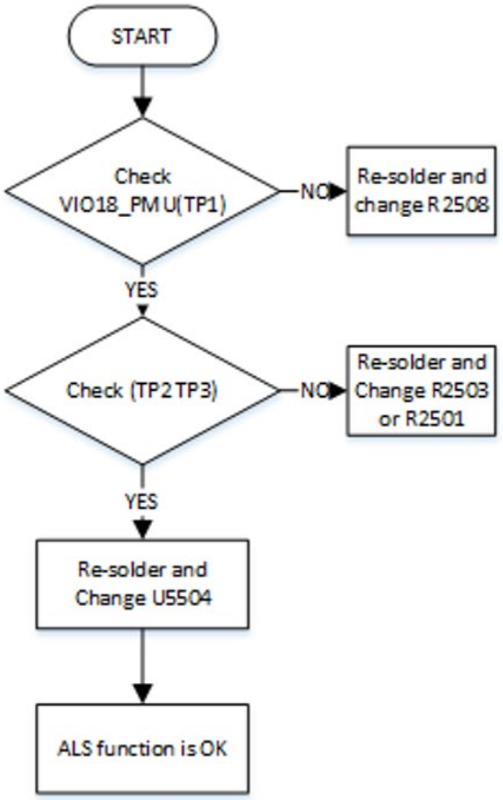
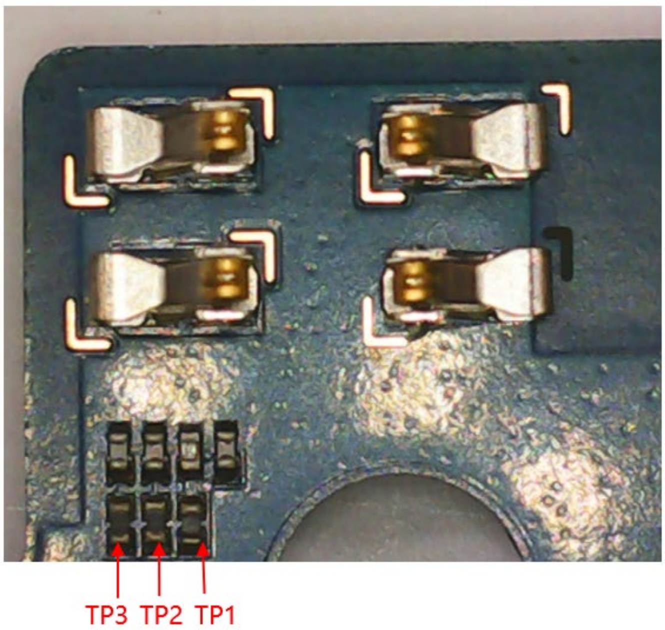
- The G sensor is calibrated by using SW algorithm.

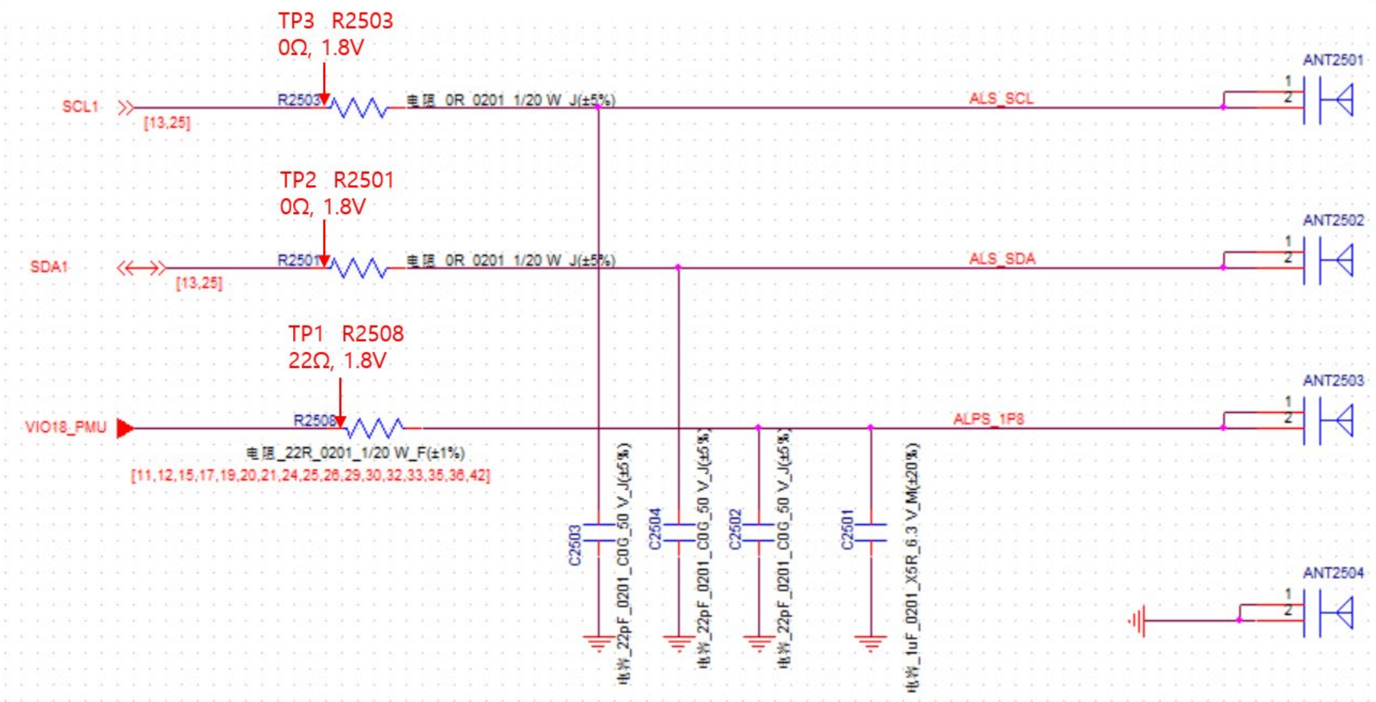
CHECKING FLOW	IMAGE
	



8-3-8. Light sensor

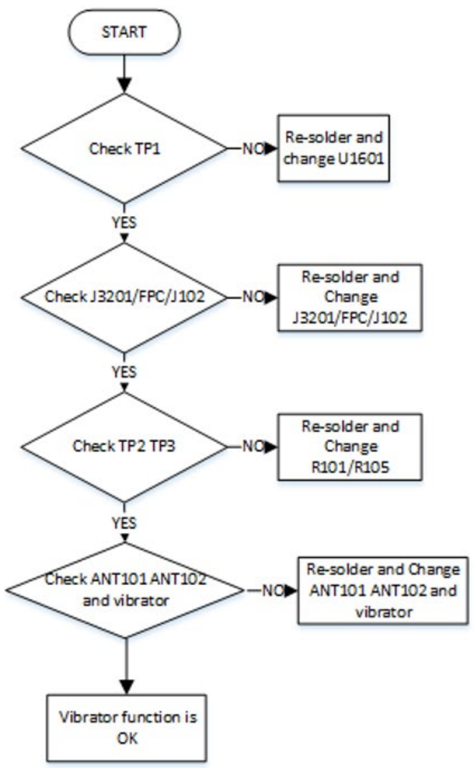
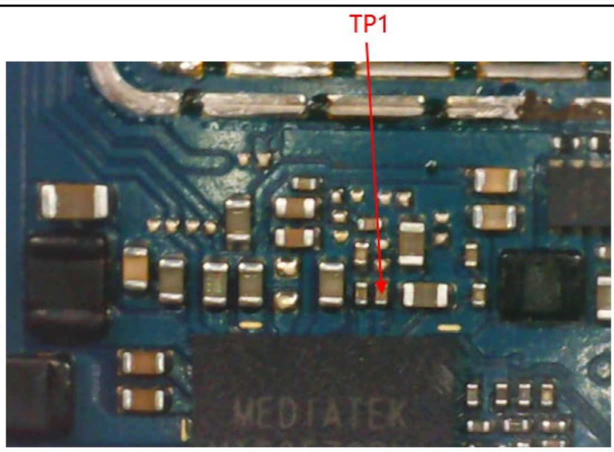
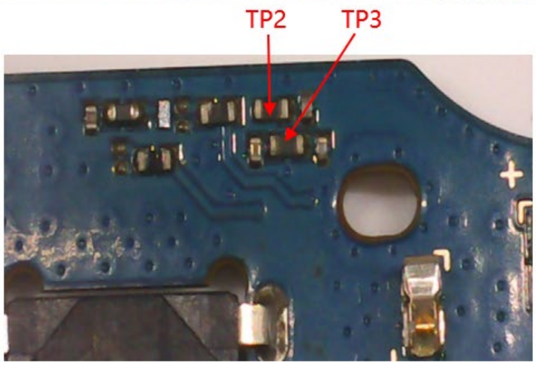
- Light Sensor is worked as below: adjust the screen brightness according to ambient light.

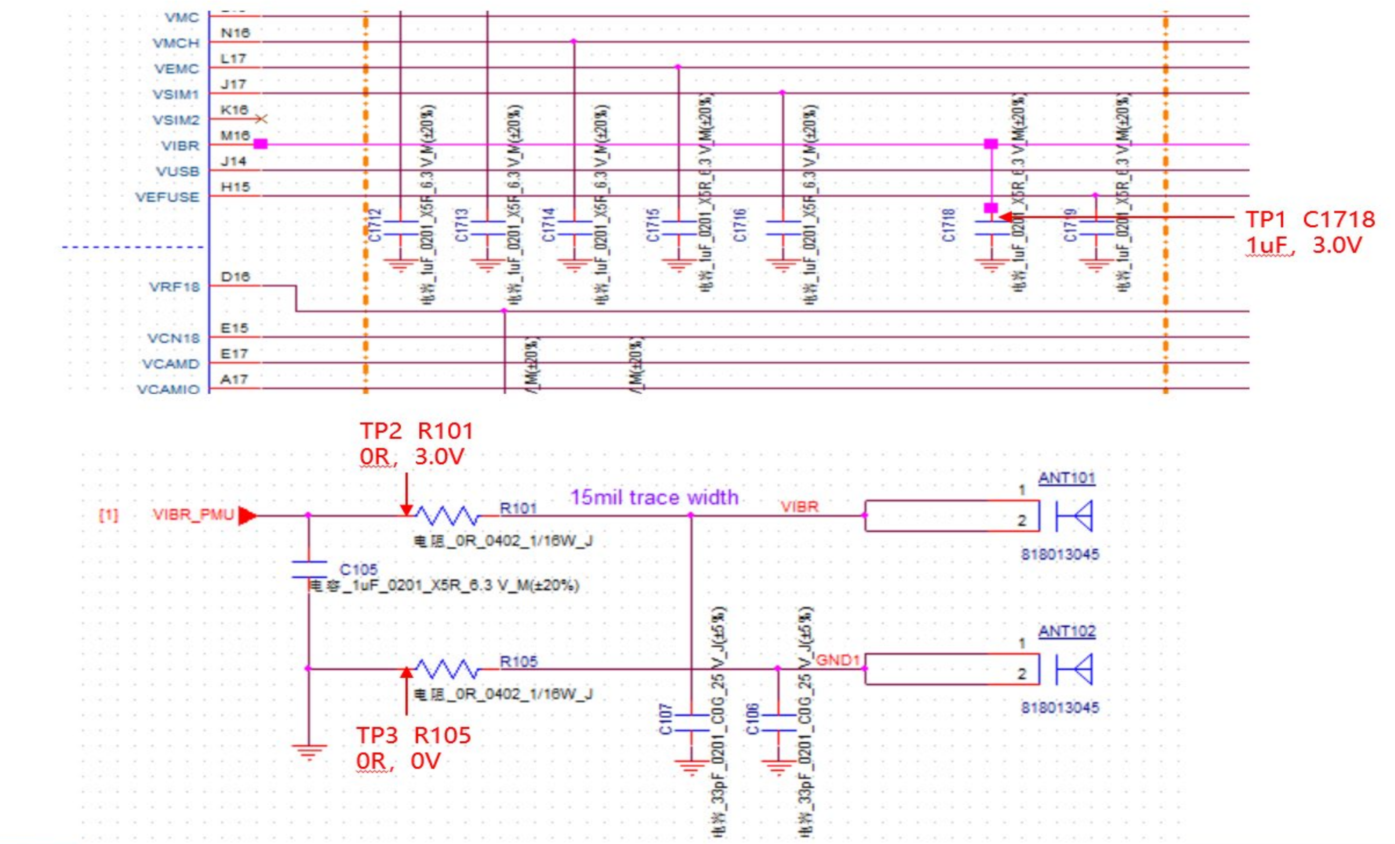
CHECKING FLOW	IMAGE
	



8-3-9. Vibrator

- The Vibrator control signals are generated by MT6357(U1601).

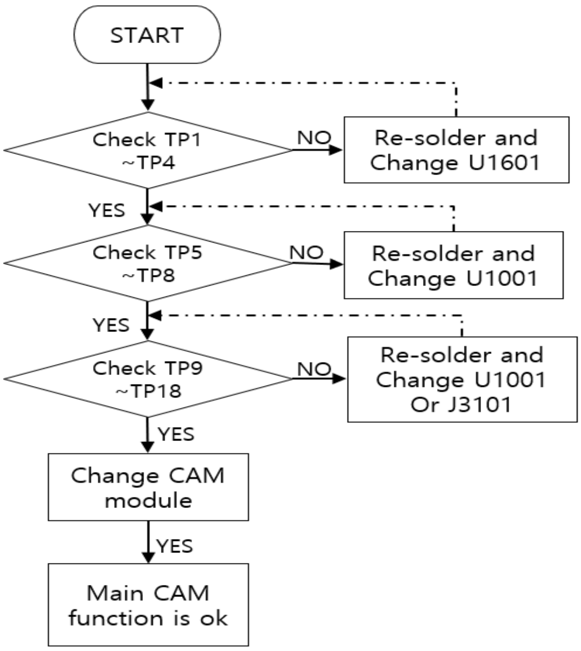
CHECKING FLOW	IMAGE
	 

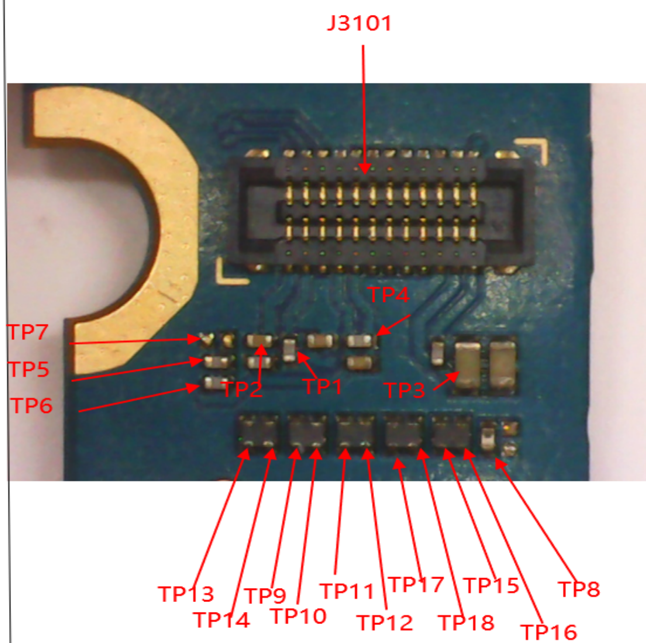


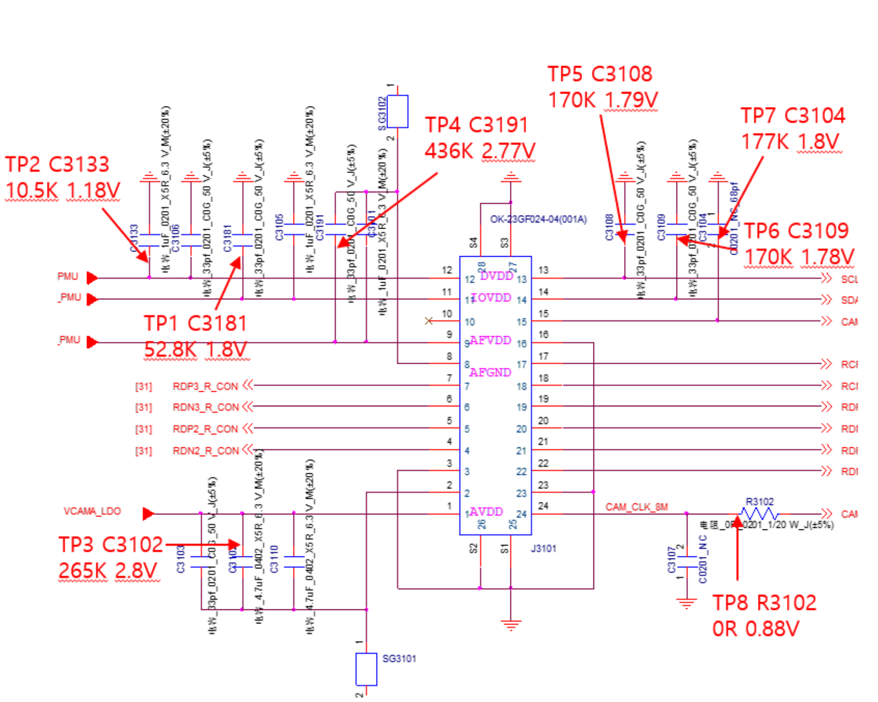
8-3-10.Main Camera

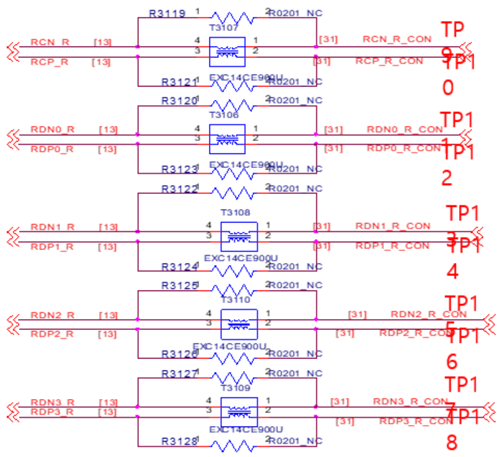
■ The Camera control signals are generated by MT6357(U1601) and MT8768 (U1001)

CHECKING FLOW	IMAGE





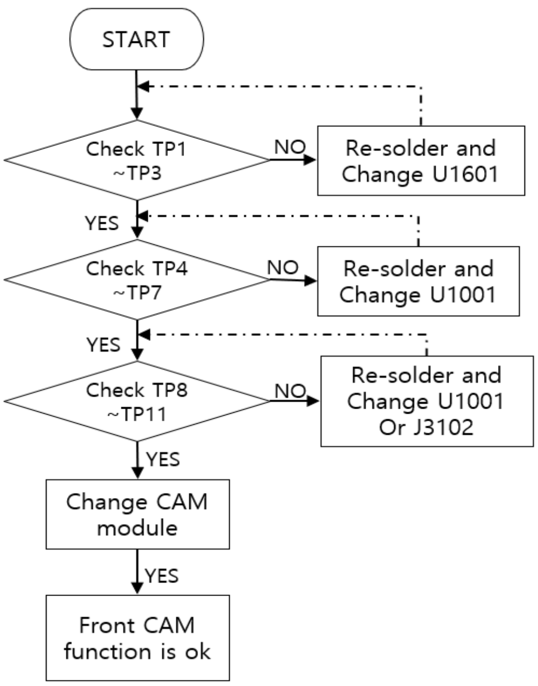
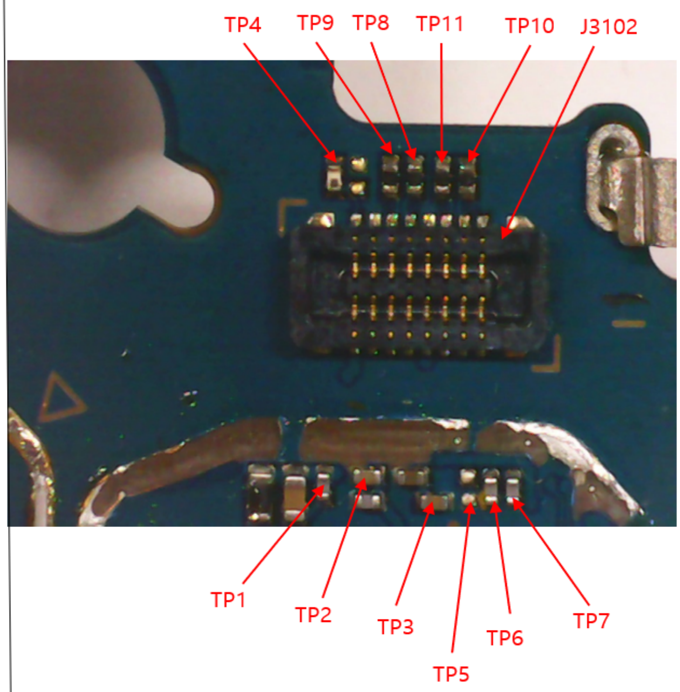




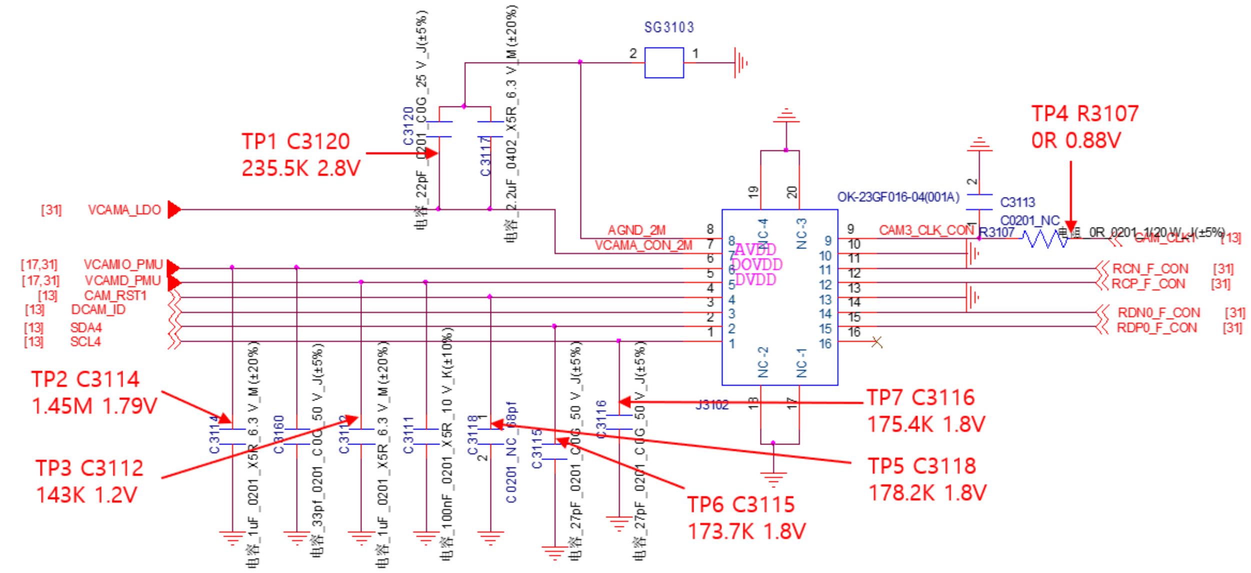
Test point	Resistance	Voltage
TP9	188.2K	0.23V
TP10	188.2K	0.24V
TP11	187.9K	0.49V
TP12	187.9K	0.44V
TP13	187.2K	0.47V
TP14	187.3K	0.43V
TP15	187.3K	0.47V
TP16	187.3K	0.43V
TP17	188.2K	0.47V
TP18	187.2K	0.44V

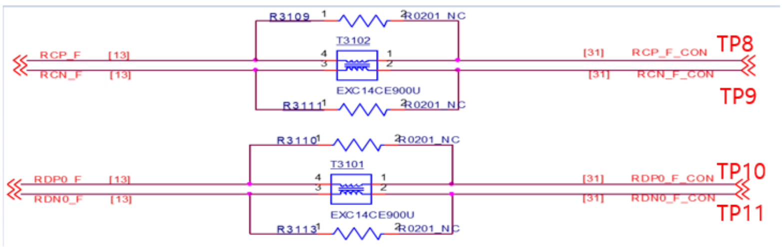
8-3-11. Front Camera

■ The Camera control signals are generated by MT6357（U1601）and MT8768 (U1001)

CHECKING FLOW	IMAGE
	

8. Level 3 Repair

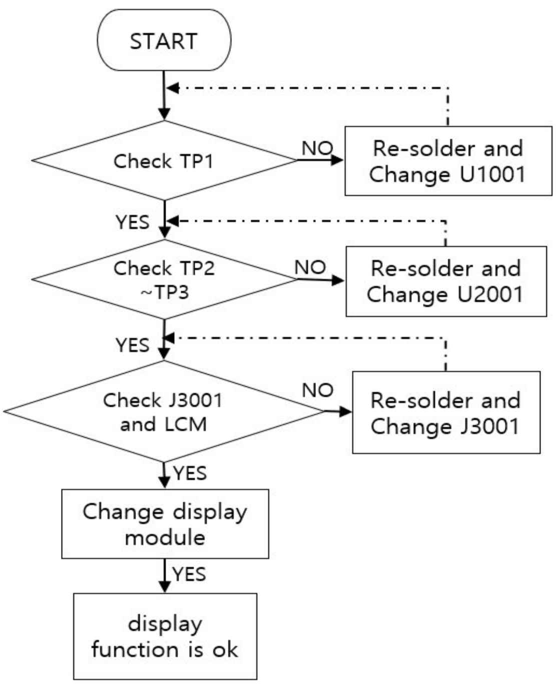
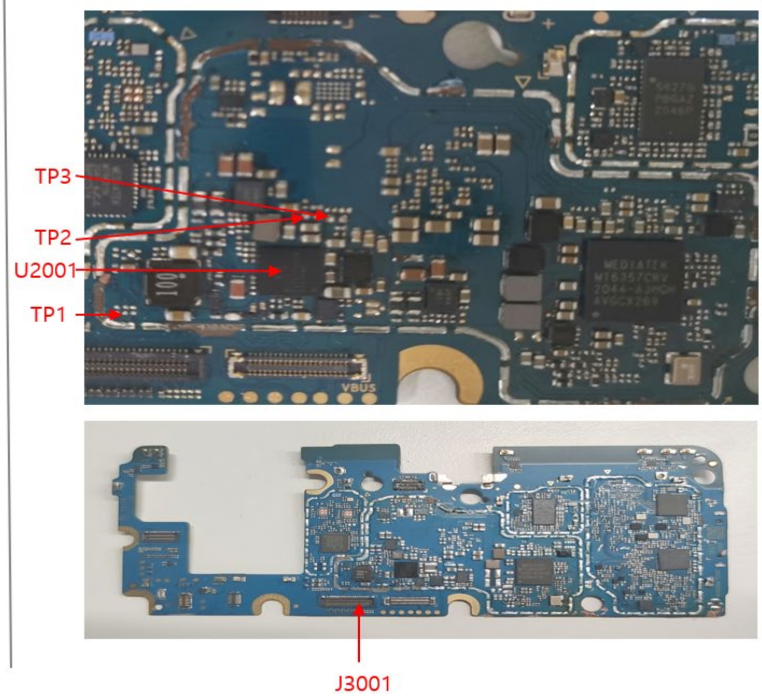


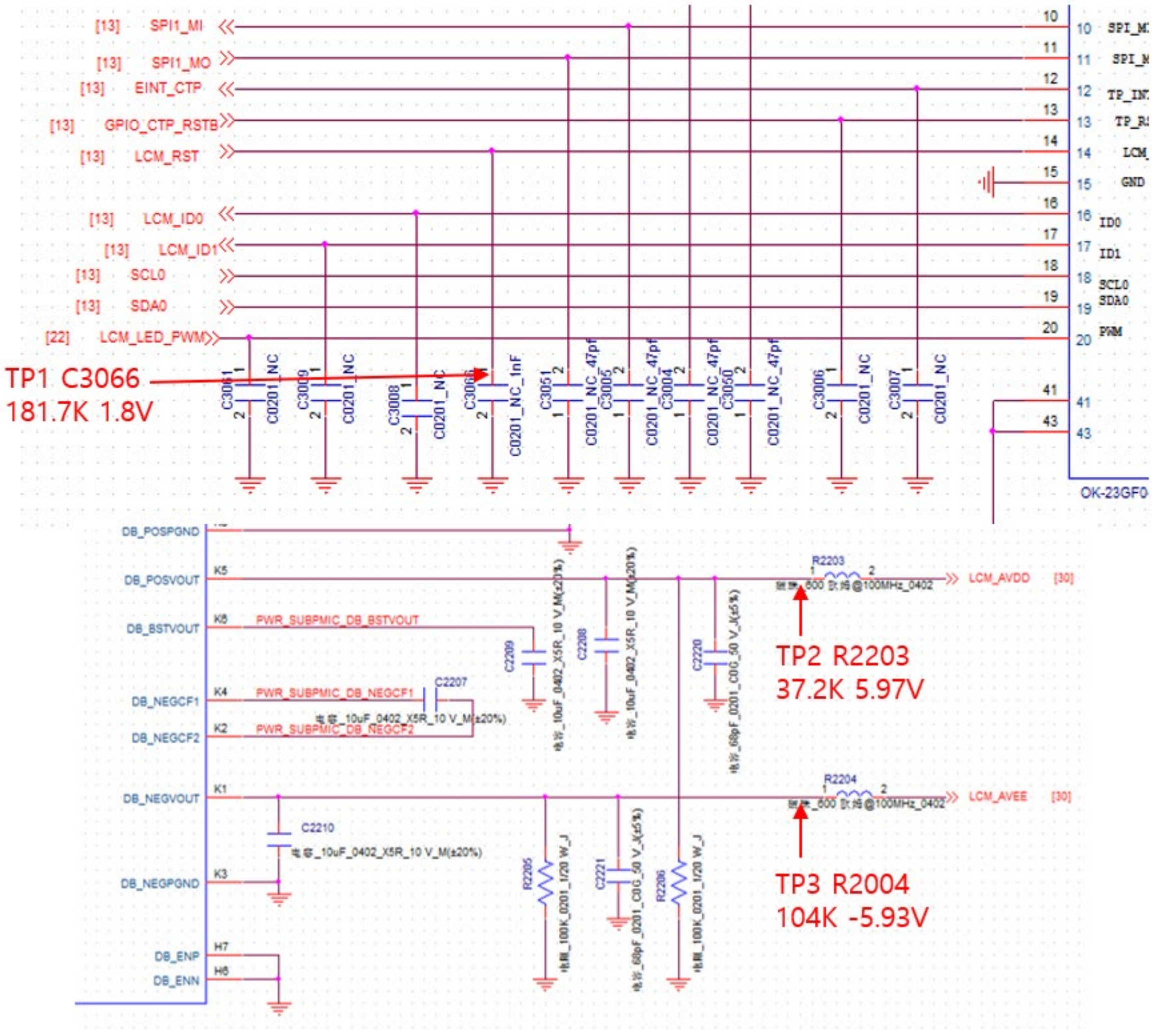


Test point	Resistance	Voltage
TP8	325K	0.28V
TP9	0.52M	0.25V
TP10	329K	0.33V
TP11	0.54M	0.38V

8-3-12. LCD

■ The LCD control signals are generated by MT8768.

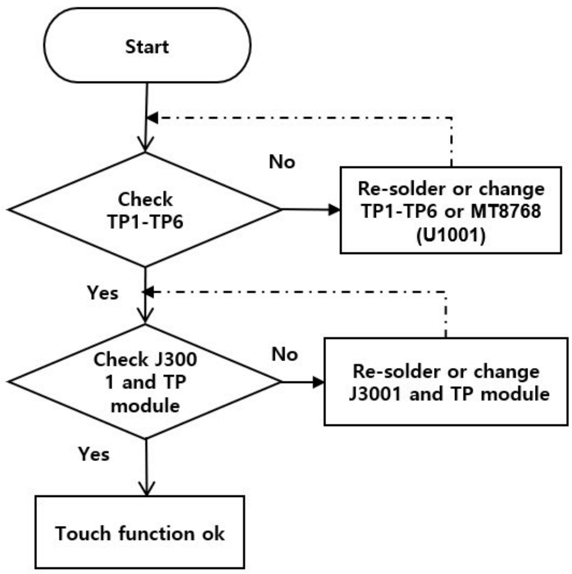
Checking Flow	Image
	


TP1 C3066
181.7K 1.8V
TP2 R2203
37.2K 5.97V
TP3 R2004
104K -5.93V

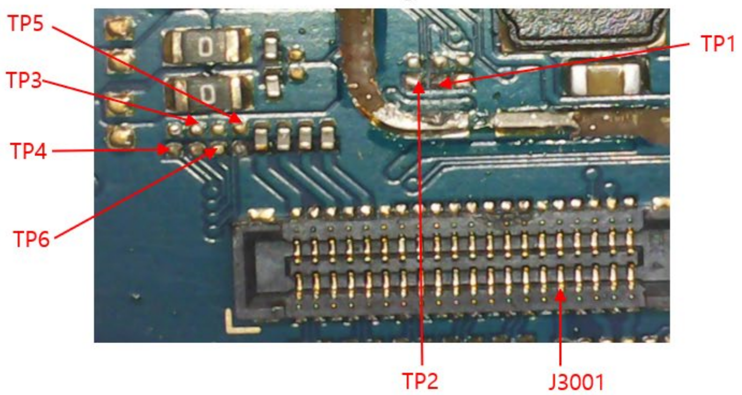
8-3-13. Touch

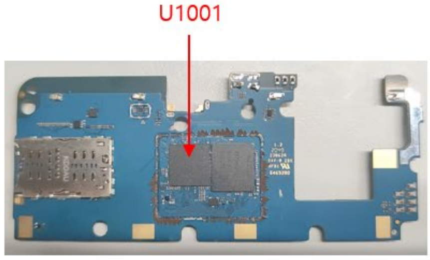
■ The Touch control signals are generated by MT8768 (U1001). It is assembled with LCD

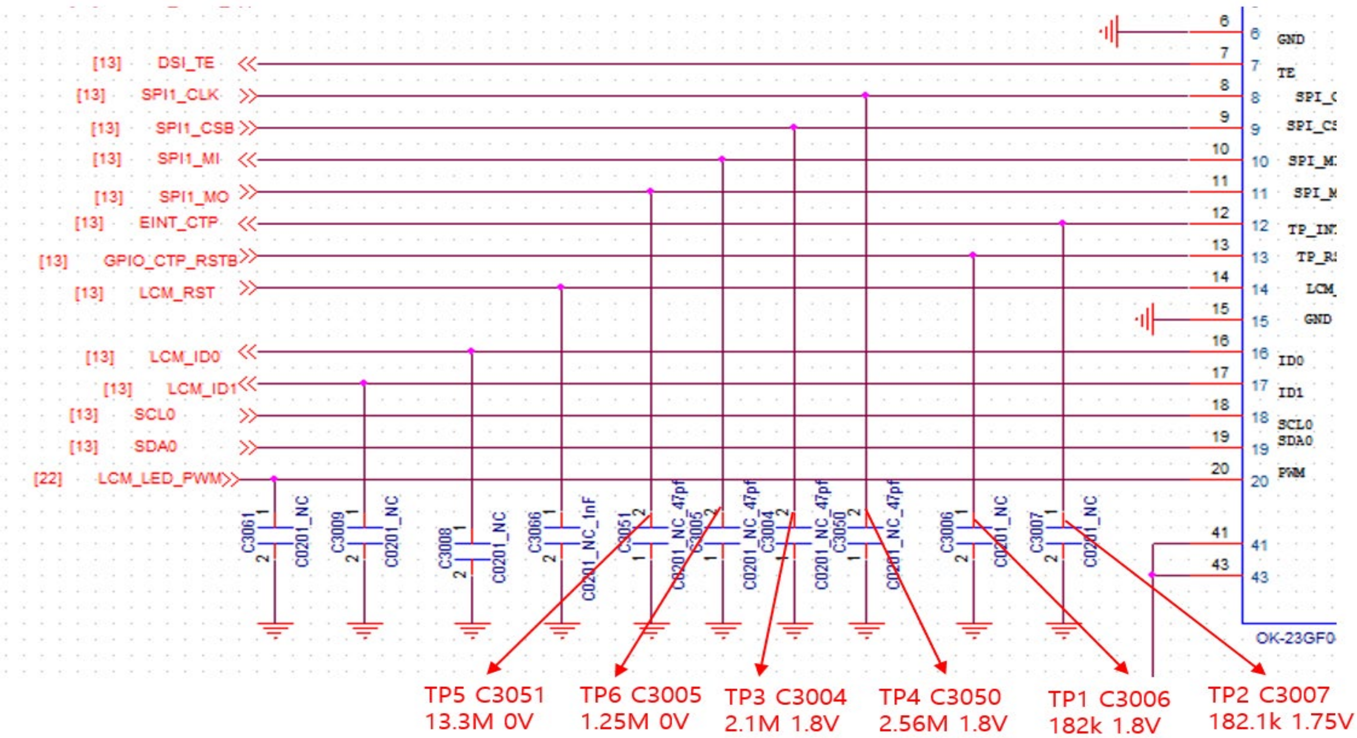
Checking Flow	Image



Image







8-4. Service Schematics

■ U1001_MT8766_BB chip IC , Ball Map view.

558	1	2	3	4	5	6	7	8	9	10	11	12	13	14
A	NC	WF_IN	WF_IP		EMI0_CA9		NC		EMI0_CS0_N		DVSS		DVSS	
B	WF_QP	WF_QN	DVSS	EMI_EXTR	EMI0_CA4		EMI0_CA1	EMI0_CA0	NC		EMI0_DQ21	EMI0_DQ22	EMI0_DQ20	
C		DVSS	AVDD12_WBG	NC	EMI0_CA6	DVSS	EMI0_CA2		NC	EMI0_DQ18	EMI0_DQ17	DVSS	EMI0_DQ0	EMI0_DQ2
D	BT_IN	BT_IP	DVSS	NC	EMI0_CA5	EMI0_CA3	NC	EMI0_CKE0	DVSS	EMI0_DQ16	NC	EMI0_DQ19	DVSS	EMI0_DQS0_T
E			BT_QP	DVSS	EMI0_CA8	EMI0_CA7	EMI0_CKE1	EMI0_CS1_N	EMI0_CK_C	EMI0_DM2	EMI0_DQ23		EMI0_DQS2_C	EMI0_DQS0_C
F	GPS_I	DVSS	BT_QN	DVSS	DVSS	DVSS	DVSS	DVSS	EMI0_CK_T	DVSS	DVSS	DVSS	EMI0_DQS2_T	DVSS
G	GPS_Q	CONN_WB_PTA	CONN_HRST_B	DVSS		DVSS	XIN_WBG				DVSS	DVSS	DVSS	DVSS
H		AVDD18_WBG	CONN_WF_CTRL2	CONN_WF_CTRL1	CONN_WF_CTRL0					AVDD2_EMI	AVDDQ_EMI0	AVDDQ_EMI0	AVDD2_EMI	DVSS
J	DVDD18_IORT	CAM_CLK2	SCL6	SDA6	CONN_BT_CLK	CONN_TOP_CLK								
K		CAM_PDN2	CAM_RST2	ANT_SEL2	ANT_SEL0	ANT_SEL1	CONN_TOP_DATA	CONN_BT_DATA	DVSS	DVDD_TOP	DVSS	DVSS	DVDD_TOP	DVDD_TOP
L	CSI1A_L1P	CSI1A_L1N	CSI1A_L0P	CSI1A_L0N	CSI1B_L0P			DVSS			DVSS		DVDD_TOP_SRAM	
M	CSI1A_L2P	CSI1A_L2N	CSI1B_L1P	CSI1B_L1N	CSI1B_L0N	DVSS	DVSS			DVDD_TOP		DVSS		DVDD_TOP
N	CSI0A_L1P	CSI0A_L1N	CSI0A_L0P	CSI0A_L0N	CSI0B_L2P	DVSS	DVSS				DVSS		DVDD_TOP	
P	CSI0A_L2P	CSI0A_L2N	CSI0B_L0P	CSI0B_L0N	CSI0B_L2N		DVSS			DVDD_TOP		DVSS	DVDD_TOP	DVDD_TOP
R	CSI2A_L0P	CSI2A_L0N	CSI0B_L1P	CSI0B_L1N	CSI2B_L1P				DVDD_VDD_MODE		DVSS		DVDD_TOP	
T	CSI2A_L1P	CSI2A_L1N	CSI2A_L2P	CSI2A_L2N	CSI2B_L1N			DVSS		DVDD_VDD_MODE		DVSS		AVDD12_PLLGP
U	DVDD18_IORB	AVDD12_CSI	CSI2B_L0P	CSI2B_L0N		SCL4			DVDD_VDD_MODE		DVSS		DVDD_VDD_MODE	
V	CAM_RST1	CAM_RST0	SDA2	SCL2		SDA4		DVSS		DVDD_VDD_MODE		DVDD_TOP_SRAM	DVDD_VDD_MODE	
W		CAM_CLK1	EINT10	EINT9	CAM_CLK0	CAM_PDN0	EINT11	DVDD_VDD_MODE	DVDD_VDD_MODE	DVDD_VDD_MODE		TN_PLLGP	TP_PLLGP	
Y	SRCLKENAI	CAM_PDN1	EINT7	EINT6		EINT4	EINT5	DVDD_VDD_MODE						
AA	KPROW0	UTXD0	URXD0	EINT3	EINT2	EINT8	EINT1		DVSS		DVSS		DVSS	
AB	KPROW1	KPCOL1	KPCOL0	EINT0	SCL0	CDM5PSA				DVSS		DVSS	MAIN_X26M_IN	
AC	SPI0_CSB	SPI0_MO	PWM0	SDA0	CDM3PSA	BPI_BUS7			RFIC_ET0_N	APC				
AD	SPI0_MI	SPI0_CLK	BPI_PA_VM0	BPI_BUS11_OLAT0	MISC_BSI_DO3	MISC_BSI_DO2	DVSS	RFIC_ET0_P	TX_BBIN0	TX_BBQN0	DVSS	DRX_BB_Q0P	DRX_BB_I0P	DVSS
AE	SCL1	BPI_PA_VM1	BPI_BUS14_ANT1	BPI_BUS0	MISC_BSI_CK3	MISC_BSI_CK2		DVSS	TX_BBIP0	TX_BBQP0	DVSS	DRX_BB_Q0N	DRX_BB_I0N	DVSS
AF	SDA1	BPI_BUS1	BPI_BUS2	BPI_BUS3	MISC_BSI_CK1	MISC_BSI_DO1	MISC_BSI_DO0	DVSS	DET_INO	DET_QPO	DVSS	PRX_BB_Q0P	PRX_BB_I0P	DVSS
AG	NC	BPI_BUS4	BPI_BUS5	BPI_BUS6		DVDD18_IORB	MISC_BSI_CK0		DET_IPO	DET_QNO		PRX_BB_Q0N	PRX_BB_I0N	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14

558	15	16	17	18	19	20	21	22	23	24	25	26	27	28	
A	DVSS		EMI0_DQ8		EMI0_DQ15		EMI0_DQ31		MSDC0_DAT4		MSDC0_DAT2	MSDC0_DAT7	DVDD18_MSDC0	NC	A
B	EMI0_DQ4	EMI0_DM0	EMI0_DQ7		EMI0_DM3	EMI0_DQ28	DVSS		MSDC0_DAT5	MSDC0_DAT6	MSDC0_CMD	MSDC0_DSL	USB_DM	AVDD33_USB	B
C	EMI0_DQ1	DVSS	EMI0_DM1	EMI0_DQ11	EMI0_DQ14	DVSS	EMI0_DQ30	EMI0_DQ29	DVSS	MSDC0_DAT3	MSDC0_RSTB	MSDC0_DAT0	USB_DP		C
D	EMI0_DQ5		EMI0_DQS1_C	NC	EMI0_DQ12	EMI0_DQ24	EMI0_DQ27	EMI0_DQS3_T	DVSS	MSDC0_DAT1	MSDC0_CLK	CHD_DP	DVSS	AVDD18_USB	D
E	EMI0_DQ3	EMI0_DQ6	EMI0_DQS1_T	EMI0_DQ9	EMI0_DQ13	EMI0_DQ25	EMI0_DQ26	EMI0_DQS3_C	NC	DVSS	DVSS	CHD_DM	SYSRSTB	AVDD12_USB	E
F		VREF_EMI	DVSS	EMI0_DQ10	DVSS	DVSS	DVSS	DVSS	DVSS	GPIO_EXT2	GPIO_EXT1	GPIO_EXT4	TESTMODE		F
G	DVSS	NC					DVSS	DVSS	GPIO_EXT0	GPIO_EXT3	GPIO_EXT5	DVSS	AUD_CLK_MISO	DVDD18_IOLT	G
H	DVSS		AVDD2_EMI	AVDDQ_EMI1	AVDDQ_EMI1	AVDD2_EMI	AVDD18_DDR	DVSS	GPIO_EXT6	GPIO_EXT7	DVSS	RTC32K_CK	AUD_SYNC_MISO		H
J							DVDD_TOP	DVSS	DVSS	DVSS	AUD_CLK_MOSI	AUD_SYNC_MOSI	AUD_DAT_MISO0	AUD_DAT_MISO1	J
K	DVSS	DVSS	DVDD_TOP	DVSS		DVDD_MFGSYS				WATCHDOG	AUD_DAT_MOSI1	AUD_DAT_MOSI0	PWRAP_SPI0_CSN	PWRAP_SPI0_MI	K
L	DVSS	DVSS	DVDD_TOP		DVDD_MFGSYS	DVDD_MFGSYS	DVSS	DVSS		CAM_PDN3	SRCLKENA0	DVSS	PWRAP_SPI0_CK		L
M		DVSS		DVDD_TOP		DVDD_MFGSYS		DVSS		DVSS	SRCLKENA1	EINT12	PWRAP_SPI0_MO	DVSS	M
N	DVSS		DVDD_TOP		DVDD_MFGSYS	DVDD_MFGSYS	DVSS	DVSS			DSI0_D0P	DSI0_D0N	DSI0_D1N	DSI0_D1P	N
P		DVSS		DVDD_TOP		DVDD_MFGSYS	DVSS	DVSS	DVDD_TOP		DSI0_CKP	DSI0_CKN	DSI0_D3P		P
R	DVSS		DVDD_TOP		DVDD_MFG_SRAM		DVSS	DVSS			DSI0_D2P	DSI0_D2N	DSI0_D3N	AVDD12_DSI	R
T		AVDD18_PLLGP		DVDD_TOP		DVDD_MFGSYS	DVDD_MFGSYS	DVSS		SPI2_CLK	SPI2_MO	SPI2_CSB	SPI4_MO	AVDD04_DSI	T
U		DVDD_DVFS		DVDD_DVFS		DVDD_DVFS		DVDD_DVFS	SPI1_CSB	SPI2_MI	SPI3_MI	SPI3_CSB	SPI4_MI		U
V	DVDD_MCUSYS_SRAM	DVSS	DVSS		DVSS		DVSS			SPI1_CLK	SPI3_CLK	SPI3_MO	SPI4_CLK	SPI4_CSB	V
W			DVSS		DVSS		DVSS			SPI1_MI	MSDC1_DAT3	MSDC1_CLK	MSDC1_CMD	DVDD18_IOLM	W
Y		DVDD_DVFS		DVDD_DVFS		DVDD_DVFS	DVDD_DVFS	DVDD_DVFS	GPIO_EXT9	SPI1_MO	SIM1_SIO	MSDC1_DAT1	MSDC1_DAT0		Y
AA		DVDD_DVFS		DVDD_DVFS		DVDD_DVFS	DVDD_DVFS	DVDD_DVFS	GPIO_EXT10	GPIO_EXT8	SIM1_SCLK	SIM1_SRST	MSDC1_DAT2	DVDD18_MSDC1	AA
AB		DVDD_DVFS		DVDD_DVFS					SCL5		SIM2_SIO	DRVBUS	IDDIG	DVDD18_SIM	AB
AC				AVDD18_CPU	GPIO_EXT15	GPIO_EXT14	SCL3	SDA3	SDA5	PERIPHERAL_EN3	SIM2_SCLK	SIM2_SRST		DVDD28_SIM2	AC
AD				DVDD_VQPS	GPIO_EXT13	GPIO_EXT12	GPIO_EXT11	PERIPHERAL_EN0	PERIPHERAL_EN2	PERIPHERAL_EN1	BPI_BUS8	LCM_RST	DISP_PWM	DVDD28_MSDC1	AD
AE			AUXIN4	AUXIN3	AUXIN2	RFIC0_BSI_D0	RFIC0_BSI_EN	PERIPHERAL_EN5	PERIPHERAL_EN4	BPI_BUS12_OLAT1	BPI_BUS13_ANT0	INT_SIM1	DSI_TE	DVDD28_SIM1	AE
AF	PRX_BB_I1	DRX_BB_Q1	DVSS	REFP	AUXIN1	AUXIN0	RFIC0_BSI_D1	RFIC0_BSI_CK	CAM_RST3	BPI_BUS15_ANT2	BPI_BUS10	BPI_BUS9	INT_SIM2	DVSS	AF
AG	PRX_BB_Q1	DRX_BB_I1		AVDD18_AP	AVDD18_MD		AVDD12_MD	RFIC0_BSI_D2		CAM_CLK3		DVSS	DVDD18_IOBL	NC	AG
	15	16	17	18	19	20	21	22	23	24	25	26	27	28	